

LEARNING
FROM
**BEST
PRACTICE
TRAPS**
IN PUBLIC
SERVICE



CONTENTS

<u>WHEN BEST PRACTICES ARE NOT BEST</u>	5
<u>LEARNING FROM THE TRAPS</u>	6
<u>TRAP 1:</u> <u>Beware of outliers that others can't replicate</u>	8
<u>TRAP 2:</u> <u>Beware of assuming consistent stakeholder behaviors</u>	12
<u>TRAP 3:</u> <u>Beware of accumulated contextual differences</u>	15
<u>LEARNING THROUGH ACTIVE DESIGN</u>	18
<u>IDEAS THAT WORK</u>	20
<u>EDUCATION IN BRAZIL –</u> <u>Sobral</u>	24
<u>URBAN PLANNING IN BRAZIL –</u> <u>Osasco</u>	27
<u>TRANSPORT IN BRAZIL –</u> <u>Curitiba</u>	30
<u>PUBLIC BUREAUCRACY IN BRAZIL –</u> <u>São Paulo</u>	34
<u>EDUCATION IN INDIA –</u> <u>Andhra Pradesh</u>	38
<u>URBAN PLANNING IN INDIA –</u> <u>Hyderabad</u>	42
<u>TRANSPORT IN INDIA –</u> <u>Delhi</u>	46
<u>PUBLIC BUREAUCRACY IN INDIA –</u> <u>Hyderabad</u>	50
<u>EDUCATION IN SOUTH AFRICA –</u> <u>Umlazi</u>	54
<u>URBAN PLANNING IN SOUTH AFRICA –</u> <u>eThekwin</u>	58
<u>TRANSPORT IN SOUTH AFRICA –</u> <u>Cape Town</u>	62
<u>PUBLIC BUREAUCRACY IN SOUTH AFRICA –</u> <u>Polokwane</u>	66

LEARNING FROM BEST-PRACTICE TRAPS IN PUBLIC SERVICE

PUBLIC SERVICE PROJECTS IN EMERGING MARKETS LOOK TO BEST-PRACTICE INITIATIVES FOR GUIDANCE, BUT OFTEN FAIL TO ACHIEVE DESIRED OUTCOMES. WHILE IT MAKES SENSE TO EMULATE EXCELLENCE, ATTEMPTING TO COPY IT WITHOUT CAREFUL PLANNING AND ADAPTATION USUALLY ENDS IN DISASTER. WE OFFER PRAGMATIC LESSONS TO LEARN FROM SUCCESSES ELSEWHERE IN A WAY THAT REMAINS RELEVANT TO LOCAL CONDITIONS.

Sometimes, following best practices can produce the worst results. We consistently see failed attempts to replicate public service projects, especially in emerging markets where “scaling” and “replication” sound like attractively simple options to resource-strapped planners.

Replication works best for easily quantifiable, low-complexity, discrete tasks in controllable environments. That’s why we see fast-food franchises all around the world, with ingredients, uniforms, processes, standards and even interior design and architecture replicated at low cost and with minimal adjustment. Public services, in contrast, are highly complex by nature, dependent on a variety of interacting stakeholders, subject to the whims of politicians and citizens, and intimately bound up with local cultural and environmental conditions that vary within national borders, let alone across them.

Our primary research across municipal projects in Brazil, India and South Africa uncovers the traps inherent in best-practice replication and offers guidance on how to learn effectively from the successful and unsuccessful experiences of others. These lessons are brought to life through 12 detailed case studies of successful projects that exemplify our recommended approaches.

ABOUT OUR RESEARCH

Accenture collaborated with Insper - a leading Latin American non-profit school in business, economics and engineering - to conduct in-depth case studies of high- and low-performing municipal public service projects across Brazil, India and South Africa.

Our research covers education, transport, public bureaucracy and urban planning initiatives and includes more than 70 detailed interviews, impact assessments and both qualitative and quantitative analysis of the dynamics that led to success and failure.

We noted the widespread impulse among senior public servants to attempt to transfer project plans wholesale from elsewhere into their own environment, including requests for our consulting and research to support these efforts. The instinct to copy successes seems logical, given the constraints on time, finance, talent, and other resources inherent to public office. And yet, we consistently see these attempts at replication fail. We determined to understand why outright replication often fails and how public project leaders can avoid these traps and plan for success.

WHEN BEST PRACTICES ARE NOT BEST

City planners for Brazil’s business capital, São Paulo, sought to emulate the successes of Chicago, Paris, Moscow and Vienna in creating an automobile-centered, radial road system. But in the process, they overlooked several local realities, like the fact that São Paulo’s steep, hilly terrain made the proposed geometric road patterns extremely challenging to build. The design also required the city to straighten several naturally meandering rivers, increasing the risk of flooding and other environmental damage. Perhaps most counterintuitively, a radial road system requires a fixed city center, while São Paulo’s economic hub has consistently shifted over time.

Consequently, the city today suffers from frequently flooded roads and some of the world’s highest levels of traffic congestion and pollution. On paper, while São Paulo’s design replicated the best-practice elements of the other cities’ plans, it was one of the worst possible solutions planners could have conceived for the local context.

São Paulo’s urban planning challenges are far from unique, but they illustrate some of the fundamental questions brought up by best-practice replication approaches, such as: What do we mean by “best”? Who decides what is best? Can a practice be the “best” over different time periods and locations?

USE WITH CAUTION: WHAT IS A “BEST PRACTICE”?

AN EVOLVING DEFINITION

The definition of “best practice” has evolved significantly over the past four decades. It was initially a benchmarking tool for achieving incremental improvements to business processes in the manufacturing industry, as popularized by Xerox in 1979. By 2009, the Encyclopedia of Management defined best practice as “...the most efficient way of doing something. The fastest method that uses the least resources (including labor and parts) to create the highest quality output...” So, the term has evolved to denote excellence.

Today, “best practice” most commonly implies leading and innovative instances of near-perfection that others should aspire towards. This is a huge shift from its lowly original ambition to raise poor performance to somewhat above-average levels.

A LACK OF CONSENSUS

The word “best” is extraordinarily broad in its definition. Does it refer to the most innovative, state-of-the-art solution? Does it mean the most appropriate solution for that environment? Is it the most efficient approach? Does it have to be all of these simultaneously? The “best” can mean any number of desirable characteristics, depending on the user’s objectives. Unsurprisingly, therefore, international institutions (like the World Bank and OECD) and leading academics have begun to distance themselves from the term “best practice.” However, in the absence of simple alternatives to convey the commonly accepted concept, the phrase continues to hold widespread appeal and usage, especially in emerging markets.

INHERENT BIASES

The word “best” is also subjective. It is important to pay attention to who defines a project as “best.” Often, a coalition of enthusiasts with inherent biases designate a project as “best practice” because it is in their interest to do so. These may be public servants who wish to build an argument for replicating the project in their jurisdiction; they may be journalists, consultants or academics looking to showcase examples; or they might be sponsors or stakeholders that wish to be associated with success. Thus, project leaders should always take care to understand why a project or activity is an example, comprehend the techniques used to prove its performance, and know why the backing evidence says what it does.

Having established some of the inherent challenges of using the best practice concept and terminology, we now look at three specific traps that can guide us towards a pragmatic way forward.



LEARNING FROM THE TRAPS

AVOID THE TRAPS		KEY RISK	KEY MITIGANT	LESSONS LEARNED
1	BEWARE OF OUTLIERS THAT OTHERS CAN'T REPLICATE	Futility of trying to replicate conditions that cannot realistically be recreated.	Never try to replicate a single instance. Always learn from multiple projects.	BUILD ON THE WISDOM OF THE CROWD Learn from patterns and dynamics across multiple projects. Proactively design virtuous cycles based on successes in multiple circumstances.
2	BEWARE OF ASSUMING CONSISTENT STAKEHOLDER BEHAVIORS	Stakeholders (eg. citizens, employees, political sponsors) can derail a project if not bought-in.	Build mechanisms to connect and communicate across all stakeholder groups.	DESIGN-IN COLLABORATION Build coalitions of shared objectives and common interests through genuine and frequent communication.
3	BEWARE OF ACCUMULATED CONTEXTUAL DIFFERENCES	The same process can result in very different outcomes due to multiple environmental differences.	Focus on desired outcomes, not on replicating specific activities. Ensure agility to evolve with circumstances.	MONITOR AND REALIGN Define context-specific outcomes, monitor progress and continually adjust plans to meet desired outcomes and local context.

TRAP 01

BEWARE OF OUTLIERS THAT OTHERS CAN'T REPLICATE

As the interpretation of the term “best practice” has edged closer toward “leading practice”, so the targets for replication have more frequently become outlier projects that symbolize aspirational excellence. Outliers are extraordinarily difficult to replicate.

These projects are typically high-profile, pet projects that aim to prove a concept. They therefore tend to have significant sponsorship and are often led by an inspirational leader and staffed by an enthusiastic workforce that is happy to commit extraordinary time and energy to the project. These conditions can propel an initial concept-proving pilot to great success.

In contrast, attempts to recreate the same project elsewhere - often with overworked, low-paid and disinterested workers - will typically fail. Replications rarely receive the same enthusiastic support and resources as the initial innovative trailblazer.

LESSON 1 BUILD ON THE WISDOM OF THE CROWD

Never try to replicate a single instance of a successful project. That success may have been an unreproducible outlier. Rather, always try to learn from multiple projects in different contexts. Project planners need to actively design their own unique model that's relevant to their circumstances, while in the process learning from the wisdom and experience of multiple other instances.

Specifically, our research reveals the importance of learning from the patterns and dynamics that are common across multiple successful projects. The in-depth case studies described in this report, offer such patterns and dynamics. The role of virtuous cycles is critical, as they illustrate how different environmental conditions and behaviors from across a complex project can work together, reinforcing one another to generate successful outcomes.

Here's a simple example of how virtuous cycles can work, illustrated in Figure 1:

- Political sponsors commission a highly-capable and experienced project leader to design a high-profile initiative.
- The project leader establishes a decentralized organizational structure that encourages and incentivizes ideas from the workforce.
- The project leader leverages her connections within and outside the public sector to find synergies and bring new knowledge and skills.
- The freedom to develop and contribute their own ideas motivates the workforce to seek a better understanding of the citizens they are trying to serve, so that their ideas can be more relevant to the communities they are serving.

- The employees feel a greater sense of ownership of the project as they feel the project values their participation. They also gain a sense of purpose and self-improvement as they learn from external experts and collaborate with public servants in other organizations.
- The citizens also develop a sense of commitment to the project, as they feel they are being listened to and their needs considered. They, in turn, more readily provide their own input and ideas for improvement.
- As these dynamics reinforce and feed off one another, and as positive outcomes emerge, the project leader builds on the current success to further delegate and encourage innovation among the team.
- The supportive project sponsors, seeing the positive outcomes, give greater autonomy and funding to the project leader, and they benefit from the political capital of being associated with a successful, innovative project.

In this illustrative case, we see the importance of combining a decentralized model with a motivated workforce that engages with its citizens. The details of the project activities remain ambiguous – we don't even know what sector it is in. What matters is how positive dynamics can reinforce desirable behaviors and conditions. When we see similar patterns of interacting conditions and behaviors succeed in different environments, we can extract lessons that help us to plan our own projects.

TAKE ACTION

Observe the successful patterns and dynamics across multiple projects that you wish to learn from. Derive lessons to proactively design your own virtuous cycles for success.

ILLUSTRATIVE VIRTUOUS CYCLES

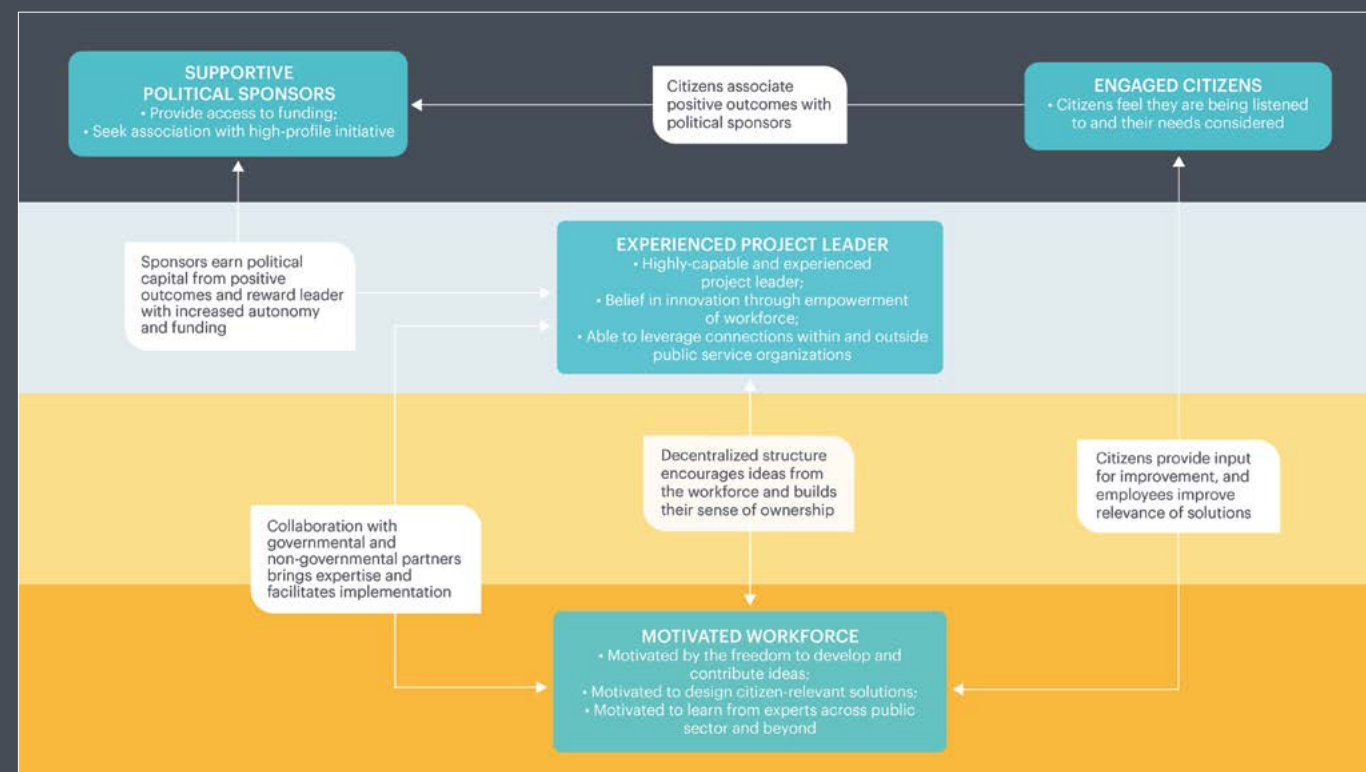
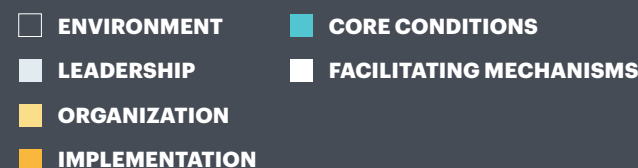


FIGURE 1
ILLUSTRATIVE VIRTUOUS CYCLE
The combination of conditions and behaviors across the political sponsors, project leader, organizational model, workforce and citizens interact to drive positive virtuous cycles.



TRAP 1 WHAT CAN GO WRONG

Astonishingly, the replication of a successful, high-profile outlier can fail - and keep failing without correction - for decades. That's decades of continually misallocated public investment. In our analysis of failed projects, we found that this kind of error can go undetected if the outlier's reputation remains intact and there is no serious monitoring of the outcomes of subsequent replications.

In one of our case studies of failure, we examined a child development initiative in service nationally since the 1970s, but overseers didn't begin to conduct the first concerted nationwide evaluation of outcomes until the 1990s. The assessment of one specific city's program took place in 2007 and uncovered dreadful findings. Less than 60% of the project's 242 facilities had toilets or access to clean drinking water, more than 80% did not have the equipment necessary for their work, 96% of workers complained of not being paid on time, and about 40% of the facilities offered poor quality food, including complaints of insects and dirt in the food. Consequent reforms implemented in 2008 had to include improvements to training, pay, record keeping, meals and equipment.

Our broader research found similar instances of long-term investment in failed replications all around the world. As President Bill Clinton noted on US school reform, "Nearly every problem has been solved by someone, somewhere." [The frustration is that] "we can't seem to replicate [those solutions] anywhere else."¹

TRAP 1 HOW TO PLAN IT RIGHT

The twelve successful case studies detailed in this report each have examples of virtuous cycles that drive mutually reinforcing positive behaviors and outcomes. Analysis of these cases teach us fundamental lessons to help design new projects:

- Projects don't succeed due to the presence or absence of individual conditions. They succeed through the dynamic interaction of multiple conditions. For instance, while a great leader may be important to success, it's how that leader interacts with the workforce, the organizational structure and other stakeholders that really drives performance.
- The interacting conditions that drive project performance spread across every aspect of an initiative and its contextual setting. This includes the environment (e.g. the institutional setting and the resources available), the leadership (e.g. the leader's ability to communicate and inspire), the organization (e.g. the degree of centralization and the nature of any collaboration), and the implementation approach (e.g. workforce skills and project management).
- By analyzing highly successful projects, we can map how different conditions and behaviors interact, thereby revealing the patterns and dynamics that drive performance. We can illustrate these interacting mechanisms and relationships through virtuous cycles.
- Rather than trying to replicate the precise dynamics of a given project, we must derive lessons from across patterns in different projects to design our own dynamics for success, relevant to our circumstances.

TRAP 02

BEWARE OF ASSUMING CONSISTENT STAKEHOLDER BEHAVIORS

Launching a new public service project involves interaction across a range of stakeholders, including citizens, public sector workers, politicians and the media. Predicting the actions and reactions of these actors is hard enough, let alone expecting their behaviors and dynamics to be similar across different contexts.

Initiatives that succeed in multiple locations can still fail in the next one because the citizens react differently to the concept or to the implementation approach. Other issues can arise because the local workforce rejects the solution in the form that worked elsewhere, or because the political establishment in the new location does not support the initiative.

Usually, these failures involve a lack of communication and the consequent lack of buy-in from stakeholders who have no sense of ownership in the project or feel it has no relevance to their circumstances.



LESSON 2 DESIGN-IN COLLABORATION

Assume very little about the behaviors and reactions of stakeholders. Instead, engage them in the planning process from the beginning. Invest in establishing formal and informal channels to connect different groups based on genuine interaction and feedback. Regular communication between stakeholders and project planners can build coalitions of shared objectives and common interest. These interactions have the added benefit of generating input to improve the local relevance of project plans. In this way, communication very naturally leads to collaboration.

The nature of the communication mechanisms will depend on what is appropriate for the context. For example, project planners may choose to connect with citizens via telephone or e-mail, or to organize focus group sessions, or sometimes the local context demands additional considerations. In one of our case studies of a project in a tribal area of South Africa, an interviewee described how they set up a big tent, brought food, and invited the community down to interact and learn from one another. This was the right way to engage the community in an environment without Internet connectivity. Sometimes, understanding the right form of communication is even more important than the details the project is communicating.

Consciously designing mechanisms for communication and collaboration is critical to building virtuous cycles. These cycles have tangible mechanisms that facilitate the interactions between stakeholders. Feedback loops are an essential part of this design, ensuring that relevant information reaches the right places. These feedback loops also allow the project design to keep up with evolving mindsets and behaviors across stakeholders, as the project progresses. Moreover, these communication mechanisms are not only a source of input to project design, they also provide a platform for project planners to inform and influence stakeholders.

TAKE ACTION

Identify the key constituents of your project and establish relevant communication mechanisms that will facilitate genuine and regular interaction across them. Feedback loops will ensure that valuable interactions translate into collaboration and tangible outcomes.

TRAP 2

WHAT CAN GO WRONG

Slum residents in one of our case studies of failed projects rejected an initiative that planned to rehouse them into apartments. A key cause of failure was that planners made faulty assumptions about the lives and expectations of the slum dwellers. They did not make sufficient efforts to understand the needs of the citizens and to plan solutions accordingly. Typically, residents lived in and worked from their slum dwellings, rooted in a well-established community that also functioned as a complex market network. So, leaving the slum dwelling also meant exiting the market network, thereby losing jobs and livelihoods. As attractive as an apartment might be, the resettled residents risked losing a lot more, hence the rejection of the re-housing solution by residents of this community. Our research unearthed multiple such instances where the very citizens that an initiative meant to serve rejected it, usually due to an absence of communication and trust.

Citizens are not the only stakeholders who can scupper project implementation. In another of our case studies, workers received GPS-enabled, digital tablet computers to improve their performance when visiting, monitoring and reporting on civil construction sites. Planners were surprised that the workforce rejected the move from a complex paper-based system to a real-time, Internet-based solution. The workers dismissed the initiative as a vanity project to show off a modern solution that was inappropriate for the deficient digital infrastructure that had to support it. There were also suggestions that members of the workforce did not welcome the capacity of the system to track their location at all times!

TRAP 2

WHAT CAN GO RIGHT

A good example of well-planned communication and collaboration mechanisms involves how eThekweni in South Africa worked with stakeholders to transform the city's water and sanitation services (see full case study on page 58). Perhaps the most interesting illustration was the recruitment and training of community liaison officers from the local unemployed population in the targeted tribal community. Their role was crucial in attempts to facilitate deep and continual communication between the planners and the citizens, who had no real access to sanitation services before the initiative. Through these interactions, the liaison officers helped plan and improve the timing, location and process of toilet installations. They made at least five visits before the installation of each facility, providing frequent opportunities for communication and learning. They also taught the citizens about the usage and maintenance of the toilets.

Beyond the interaction with citizens, this project set up intense communication and collaboration across a variety of players. The municipal authorities collaborated with other cities to develop their collective engineering skills; they worked with water research organizations to develop a national benchmark initiative; they collaborated with a local university to innovate processes and to learn about social aspects of sanitation in local communities. They also successfully partnered with the private sector for much of the construction effort; they worked with professional bodies to bring training and advice to the workforce; and they partnered with the Bill Gates Foundation on related research. All these connections drove incredible learning and innovation.

TRAP 03

BEWARE OF ACCUMULATED CONTEXTUAL DIFFERENCES

The contextual environment in two separate locations will vary according to multiple differences in the political, social, cultural, infrastructural, geographical and environmental conditions. These differences - even if small in isolation - accumulate to influence project delivery on the ground. Implementing identical processes in two locations is therefore likely to result in very different outcomes in each location.

Projects aimed at replicating successful information and communications technology (ICT) solutions offer classic examples of this dynamic. ICT projects should scale up easily, cheaply and rapidly due to the nature of the technology. But scaling is frequently hampered by a combination of environmental factors that vary across locations. Incompatible infrastructure (e.g. electricity supplies or sockets), climatic conditions (e.g. heat or dust), or weak content relevance to the local community (e.g. language or social environment) are typical examples of factors that frustrate replication attempts.

A World Bank report looking at ICT for development projects between 2003 and 2010 found a 70 percent failure rate in projects aiming to deliver universal access to ICT. In the 2000s, India gained a reputation in this field as "a graveyard of successful pilots," due to this struggle to replicate ICT project successes. More recently, city planners have been discussing why it is proving so difficult to replicate the success of "smart city" solutions across different locations.

Experience shows that the accumulated contextual differences between locations renders exact replication across multiple locations practically impossible.

LESSON 3

MONITOR AND REALIGN



Focus on outcomes, not on replicating specific activities. As Jeffrey Bradach of the Bridgespan Group, who has published, taught and advised extensively on scaling success in the non-profit sector, put it, “The objective is to reproduce a successful program’s results, not to slavishly recreate every one of its features.”²

The desired outcomes must be context-specific. Furthermore, there must be mechanisms to monitor progress toward those outcomes, and to bring attention to situations where there is divergence from them. This monitoring and continual realignment makes the project a living, breathing part of its context. It ensures local relevance even as circumstances and behaviors evolve during the project’s progress.

Projects are dependent on communication channels and feedback loops to support the reporting of monitored information. This is how the information moves to where it is needed. These mechanisms can warn planners of emerging issues, inform sponsors of progress, allow citizens to report their grievances and let employees see the impact of their work. Monitoring and communication are intrinsically connected and their role in a project should be designed together.

TAKE ACTION

Define context-specific outcomes and establish monitoring and reporting mechanisms to measure progress toward those outcomes. Use regular reporting to continually adjust to the local context and target outcomes.

TRAP 3

WHAT CAN GO WRONG

We investigated a failed attempt to implement a school cyber-lab, one of 186 labs implemented by the same organization in that year. But experience is no guarantee of success.

Cyber labs are relatively common in the target country and the implementing organization had significant recent experience, so in theory, this should not have been a difficult concept to replicate. The project leader had good intentions, bringing together players from different sectors and securing the necessary equipment. However, the initiative never articulated its target outcomes and failed to share an implementation plan with the project participants; not even with the school leadership and participating teachers. The sheer number of hurdles presented by their environment, including deficits in skills, infrastructure and communication, accumulated to seal the project’s fate. The tangible outcomes that the project sought to achieve failed to materialize, and in their place the only results consisted of a set of “best practice” activities on a checklist. The project also neglected to monitor for ongoing implementation issues.

One of the interviewees described to us, “a beautiful picture, seeing all the computers in the lab...it’s a shame we never got to use them.” And one of the planners admitted that such initiatives often struggle due to the lack of ongoing maintenance of the machines. But that was just one piece of straw that helped break the camel’s back. Most of the teachers lacked the training to use the computers, let alone implement software updates to keep them operational. Even the relevant representatives from the national department of education could not support the system; they were unable to use e-mail and were only contactable by phone. The ecosystem of schools, private companies and government departments failed to collaborate or even communicate to make the system work for the community. In fact, the school director told us that she didn’t realize they had access to the internet until two years later, when the telecom company came to cut the connection. They didn’t even know that the firm that donated the computers had paid for the internet access.

TRAP 3

WHAT CAN GO RIGHT

Hyderabad in India implemented eSeva, an Internet-enabled one-stop shop for citizens to access services like utility and tax payments as well as popular business services such as mobile phone payments (see full case study on page 50). Monitoring mechanisms played a key role in the project’s success. A dedicated center was established to monitor performance through key performance indicators, budgetary controls and quality assurance audits. Data on service usage was regularly backed up to ensure system continuity and the center utilized remote security cameras to observe the day-to-day functioning of the eSeva electronic kiosks.

The impact of this monitoring was profound. Most important, the information gathering allowed managers to improve the project’s design continually, drawing input from a variety of sources, not least from the observed behaviors and input of citizens. In fact, the project’s success depended on the uptake of the system by citizens, which meant making the model as customer-friendly as possible. The kiosks increased transparency, clearly displaying prices for services, for instance, and kiosk personnel received regular customer service training. The project quickly gained a reputation for being highly sensitive to user feedback, and citizens saw the system improve in relevance and convenience, thanks to their input and monitored observations. For example, in more rural zones, kiosks were adapted to be more relevant by providing both digital and manual services, enabling users to print out and “rubber stamp” documents, resulting in higher solution adoption rates.

Monitoring also allowed effective and regular reporting to sponsors and stakeholders, keeping them engaged and supportive. It was also an important tool in minimizing corruption, and the data and lessons learned from eSeva also helped to improve other public service initiatives and plans.

The eSeva initiative illustrates how monitoring is not only about accountability, its impact is widespread, improving citizen satisfaction and ensuring continual relevance, even as conditions evolve.

LEARNING THROUGH ACTIVE DESIGN

OUR ANALYSIS OF BEST-PRACTICE TRAPS OFFERS A PRAGMATIC FIVE-STEP APPROACH TO LEARNING FROM THE PAST PROJECT EXPERIENCES OF OTHERS.

The lessons learned from our analysis of best practice traps demand an active approach to project design. “Best practice” approaches are too often seen as a short cut that can save hard work and planning. But project success is impossible without project relevance. That relevance demands active design and continuous adaptation by those that understand the local context. We offer five steps, derived from our analysis of success and failure, to support public service leaders along this path.

Our research also suggests that few public service institutions in emerging economies develop the leadership skills or organizational capabilities required to undertake this kind of active design. Collaboration with the private sector, civil society or other parts of the public sector can, to some extent, compensate for skills gaps. But it is usually the expertise of the top planning team that stimulates the positive dynamics of virtuous cycles. The more capable and experienced the leadership, the better the starting point, the smoother the coordination, and the greater the ability to continuously learn and evolve across the entire project. We therefore urge public service institutions to build these capabilities among their leaders, most crucially, the ability to learn continuously from the successes and failures of others.

The following five pragmatic steps can help planners actively design projects that learn from the experience of others, while ensuring relevance to their citizens:

- 1 CAREFULLY SELECT YOUR TEACHERS**
Be clear about why you choose a successful case to learn from. Focus on finding projects that achieved outcomes similar to your aims. Pay attention to the evidence that shows those outcomes were achieved.

For example, are you able to find counterfactual data about what would have happened if the project were not implemented? Could the sources of the evidence be biased?
- 2 BUILD ON THE WISDOM OF THE CROWD**
Ensure that you have multiple successful cases to learn from. Learning from patterns and dynamics that succeeded in a variety of contexts reduces the risk of trying to replicate unreproducible outliers.
- 3 ACTIVELY DESIGN VIRTUOUS CYCLES**
Learn from the successful patterns and dynamics of other projects to actively design your own virtuous cycles, relevant to your resources, desired outcomes and context.

Consider the holistic set of conditions impacting your project, including the external environment, leadership, organization and implementation.
- 4 DESIGN-IN COLLABORATION**
Incorporate mechanisms to connect stakeholders and facilitate genuine and regular interaction when designing virtuous cycles.

Ensure that the communication methods are relevant to the stakeholders and construct feedback loops to transmit key information from those interactions into the planning process. The goal of this process is for communication to naturally transform into collaboration.
- 5 MONITOR AND REALIGN**
Establish monitoring mechanisms to ensure the project progresses in alignment with target outcomes and local needs. It is impossible to predict the millions of small ways in which a project’s local context will force it to diverge from the initial plan. A continual process of monitoring and realignment should also be supported by the feedback loops (described above) to create a positive dynamic of context-driven adaptation.

IDEAS THAT WORK

WE PRESENT **12 IN-DEPTH CASE STUDIES** OF HIGHLY SUCCESSFUL MUNICIPAL PROJECTS IN **BRAZIL, INDIA AND SOUTH AFRICA.**

These case studies bring to life the core concepts we have introduced in this report:

- They illustrate how feedback loops and virtuous cycles can power the dynamics of success;
- They highlight the importance of effective communication mechanisms to drive insight, collaboration and alignment; and
- They underline the role of monitoring and continuous adjustment to maintain local relevance.

Each case study includes an introduction of the project context, its impact and its challenges.

This is followed by a detailed description of the key dynamics that drove success, including a description of the project conditions that were most central to achieving high performance.

Each case study description includes a diagram illustrating the project’s performance dynamics. These diagrams illustrate how the key project conditions interact with one another through mechanisms that generate feedback loops and virtuous cycles.

NOTE

Our fieldwork research explicitly avoided the terminology of best practice. Moreover, we took important steps to ensure the objectivity and rigor of our assessments of project performance. For example:

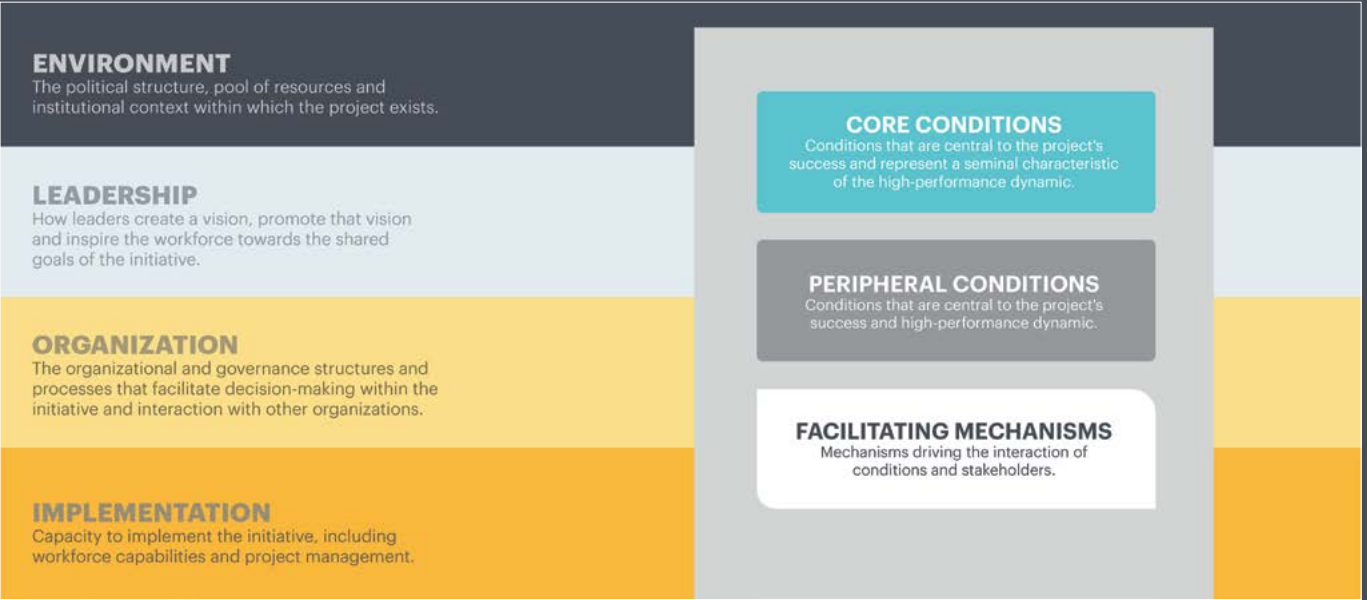
- We evaluated the high-performer projects based on objective, evidence-based outcomes, measured against counterfactual data from third-party impact assessment studies.
- We conducted multiple interviews for each case study, bringing different perspectives, which we then tested for statistical consistency.
- A robust interview protocol ensured the assessment of the widest possible range of project conditions, allowing a holistic view of the circumstances of each project.

OUR TWELVE HIGH PERFORMER CASE STUDIES

FROM ACROSS BRAZIL, INDIA AND SOUTH AFRICA EXEMPLIFY HOW THESE VIRTUOUS-CYCLE DYNAMICS CAN WORK IN PRACTICE

HIGH-PERFORMER CASE STUDIES	BRAZIL	INDIA	SOUTH AFRICA
EDUCATION	Sobral Literacy Program Sobral / Ceará	Teacher Performance Pay 5 Districts / Andhra Pradesh	Siyakha Nentsha Umlazi / KwaZulu-Natal
URBAN PLANNING	Land Titling Regularization Osasco / São Paulo	Water Supply and Sewerage Board Hyderabad / Andhra Pradesh	Water and Sanitation Supply eThekweni / KwaZulu-Natal
TRANSPORT	Bus Rapid Transit (BRT) Curitiba / Paraná	Delhi Metro Delhi / Delhi	Bus Rapid Transit (BRT) MyCiTi City of Cape Town / Western Cape
PUBLIC BUREAUCRACY	Poupatempo São Paulo / São Paulo	eSeva Hyderabad / Andhra Pradesh	Land Titling Program Polokwane / Limpopo

KEY FOR CASE STUDY DIAGRAMS



READ ALL ABOUT IT: BOOSTING LITERACY IN BRAZIL

BRINGING LITERACY TO SCHOOL IN SOBRAL TOOK A STRONG LEADER, SOLID POLITICAL SUPPORT AND A TEACHING STAFF THAT HAD ROOM TO EXPERIMENT.

Sobral, Brazil was spending over a quarter of its budget on the education of more than 30,000 students, but its return on investment continually earned failing grades.³ In 2000, the school system had held back nearly 60% of Sobral students because they failed to achieve passing grades. Furthermore, a standardized test revealed that almost half of all students in the third year of school could not read.⁴

SHARING NEW LESSONS

Desperate to reverse this trend, Sobral's administrators launched an innovative literacy program, allowing the relocation of principals and teachers across the public school system. Such relocations are rare in Brazil, but planners saw the opportunity to spread high-performance practices across the city's schools while also empowering teachers to apply new methodologies in their classrooms. It was a huge success. A propensity score matching technique was used to compute the impact of the program. The approach emphasized the evolution of Sobral's primary and secondary school students compared to similar students from all over Brazil via standardized tests. This assessment measured the dramatic improvement in Sobral's education outcomes compared to what would have been achieved without the new program. The results revealed that Sobral had evolved substantially, reaching 97% literacy for first-year students. Moreover, Sobral not only improved education literacy, but it did so without increasing its educational system costs.⁵

AN ENDURING LEGACY

The literacy project continues to produce stellar results. For example, in 2014 Sobral had only 2% of its students in the wrong schooling year and only 4% did not know how to read.⁴ Moreover, while focused on literacy, the program has had beneficial effects on other subjects as well, perhaps because reading comprehension increased dramatically. For instance, Sobral went from being one of the lowest-ranked in mathematics and Portuguese language studies to become one of the country's best performing school systems on these subjects. The program's success helped to propel its leader from being mayor of Sobral to becoming State Governor, enabling him to spread the lessons from this initiative across the state, with similar positive results.

OVERCOMING DISTRUST AND COMPLACENCY

Situated in one of the poorest areas of northeastern Brazil, communities in Sobral would often overlook low education scores to focus on the more fundamental challenges life presented. This complacency also carried an edge of distrust, since many in the community viewed government strategies seeking greater efficiency as little more than ways to cut costs, not something that would help them and their children. Consequently, some local organizations, including unions, attempted to undercut the project's progress. Beyond this distrust, the initiative called on participating teachers and students to take on new and different responsibilities and tasks, exposing a clear lack of managerial skills. It was thus necessary to develop an entirely new program to support teacher development, although even today, training and attracting teachers with greater capabilities remains a challenge for Sobral.

HOW THEY DID IT

Based on our primary research, including interviews with a variety of stakeholders and participants, the Sobral literacy project relied on the interaction between three core drivers and two supporting ones. The program's empowered and committed leader worked with a collaborative project design and management approach and with an empowered and committed workforce. Strong support from the political and institutional backers and affected citizens reinforced these positive dynamics (Figure 2).

Sobral, Brazil

EDUCATION SOBRAL LITERACY PROGRAM

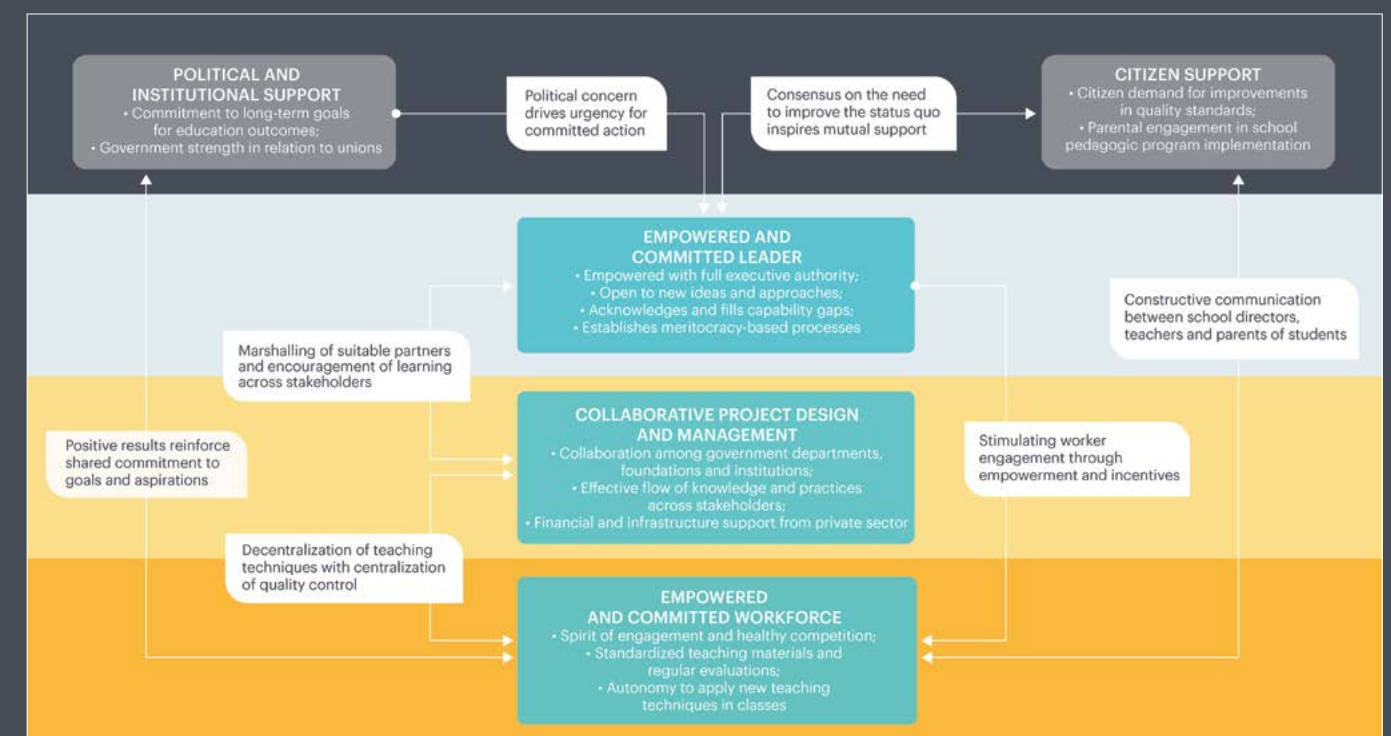


FIGURE 2

The core (blue) and peripheral (grey) drivers of Sobral's successful literacy program.

- ENVIRONMENT
- LEADERSHIP
- ORGANIZATION
- IMPLEMENTATION
- CORE CONDITIONS
- PERIPHERAL CONDITIONS
- FACILITATING MECHANISMS



CORE DRIVER 1 AN EMPOWERED AND COMMITTED LEADER

After the publication of a diagnosis exposing the precarious situation of Sobral's education system, the city's electorate empowered then-mayor Cid Gomes to lead the transformation of Sobral's literacy efforts. He received total autonomy to act, and developed a merit-based approach to select and promote teachers and directors based on their individual performance. At the same time, Gomes remained open to new ideas and methodologies that had demonstrated improvements to student literacy. This openness permeated the project, enabling it to receive support from external expertise in both project design and management.

CORE DRIVER 2 COLLABORATIVE PROJECT DESIGN AND MANAGEMENT

The project planners collaborated with the Ayrton Senna Institute and the Alfa and Beto Institute in 2000 on a program with two primary goals. The first was to achieve full literacy among school children up to seven years of age. The second goal was to promote literacy in 100% of the children who had already passed the first grade. These institutes had an in-depth knowledge of education practices and how to improve literacy conditions. They helped Sobral's education department with the development of centralized quality control and decentralized teaching methodologies. They also provided educational materials and methods, along with useful techniques for keeping students on track. Other assistance included training on both teaching and school management. This support and empowerment of the principals and teachers was crucial when it came to motivating them to improve performance.

CORE DRIVER 3 WORKFORCE EMPOWERMENT AND COMMITMENT

The involvement of the education institutes and the charismatic leadership of the mayor inspired strong engagement and commitment from the teachers and principals. In fact, this was enough to overcome the hurdles imposed by local teachers' unions. The project employed a "loose/tight" philosophy: on the one hand, it centralized quality control, with teachers instructed to use standardized teaching materials. On the other, it encouraged the use of new teaching techniques, with the only restriction being that they produce results. The project also developed feedback mechanisms to share effective ideas across the school system. The result was a sense of autonomy and empowerment among the workforce, and even a healthy sense of competition as high performers received recognition for their innovative techniques.

PERIPHERAL DRIVERS CITIZEN, POLITICAL AND INSTITUTIONAL SUPPORT

The urgency to fix a broken system was clear among stakeholders, from political leaders to the teachers and principals and the parents of the students. The mutually shared objective allowed for constructive communication across these groups. For example, parents actively participated in the implementation of the program through school board meetings. Their support and involvement was also required to enforce new stipulations around absenteeism. These and other interactions built citizen support for the program and reinforced the leader's power to improve the literacy project. What's more, the new solutions and practices developed by the teachers brought quality improvements that strengthened the project's political and institutional support. This stronger support enabled the government to challenge the teachers' unions regarding the technical criteria for the promotion and selection of teachers. These criteria normally take seniority and political considerations into account, not the merit-based factors the project used. Finally, the project tied this improvement in quality to the long-term goals desired by the citizens of Sobral, which reinforced the leader's empowerment.

GOOD DEEDS, BETTER LIVING: BRAZILIAN CITY STAGES A LAND TITLE REVOLUTION

DENSELY POPULATED OSASCO, BRAZIL, NEEDED TO TRANSFORM ITS GROWING SLUMS, WHICH MEANT A COLLABORATIVE EFFORT TO SORT OUT LAND TITLES.

Owning a home is a dream that's becoming a reality for more residents of Osasco's densely packed slums, thanks to an innovative land titling project. Located in the metropolitan area of São Paulo, Osasco is an independent Brazilian city that faced issues with high levels of immigration and overall demographic growth. In 2010, the city ranked fourth in the state in terms of population density, with 10,264 inhabitants per square kilometer.

TURNING SLUMS INTO HOMES

By 2005, Osasco housed 200,000 people in 167 slum areas that lacked proper urban infrastructure and adequate public services. That same year, Osasco adopted a new public policy for housing and urban development that included a land title regularization program.⁶ The program had four aims:

- Recognize property rights in slum areas;
- Provide legal and appropriate housing to the residents of informal slum settlements;
- Integrate the settlements into a formal urban system;
- Ensure social assistance for the communities and guidance on the importance of land ownership.

BRAINCHILD OF A NEW ADMINISTRATION

The land title regularization program was the brainchild of the new local government administration that won election in 2004. The new administration immediately began to modernize the city's housing and urban development department, installing new IT systems, building geo-referenced databases and hiring skilled staff. It quickly reformed legislation concerning urban planning and housing, involved citizens in the decision-making process by creating the Municipality Council for Housing and Urban Development, and organized forums in each targeted community to explain the transformation taking place and to prepare the affected citizens for the changes. To fund the project, the city struck partnerships with federal government agencies, and by 2012 had granted formal land titles for more than 13,000 households.⁷

In 2007, the city won the "Selo Cidade Cidadã" (literally, Citizen City Seal) award given by the Commission of Urban Development by the Chamber of Deputies for the progress made.⁸ What's more, the project's social impact evaluation revealed that the local labor force was able to work more hours as time was saved through the new streamlined bureaucracy.⁹

JUGGLING PROJECTS, WORKING WITHOUT CLEAR GOALS, LACKING INCENTIVES

The project suffered from a general lack of formal procedures concerning goals, budgets and timelines. The initiative did not have a dedicated workforce, so employees had to juggle other projects and administrative processes at the same time. In fact, team motivation and progress relied on the intrinsic drive of individual workers as there was no financial incentive system.

These conditions could have jeopardized performance, and it is a testament to the project's leadership that an effective and committed ecosystem was built, involving employees and citizens, all inspired by the same vision.

HOW THEY DID IT

Osasco's success in the land titling initiative involved the interaction of three core drivers and two peripheral ones (Figure 3). Core elements included having an experienced and expert leader, developing a collaborative ecosystem and assembling a committed and capable workforce. The two peripheral factors were the project's citizen-centered design and the existence of an enabling institutional environment.

Osasco, Brazil

URBAN PLANNING LAND TITLING REGULARIZATION

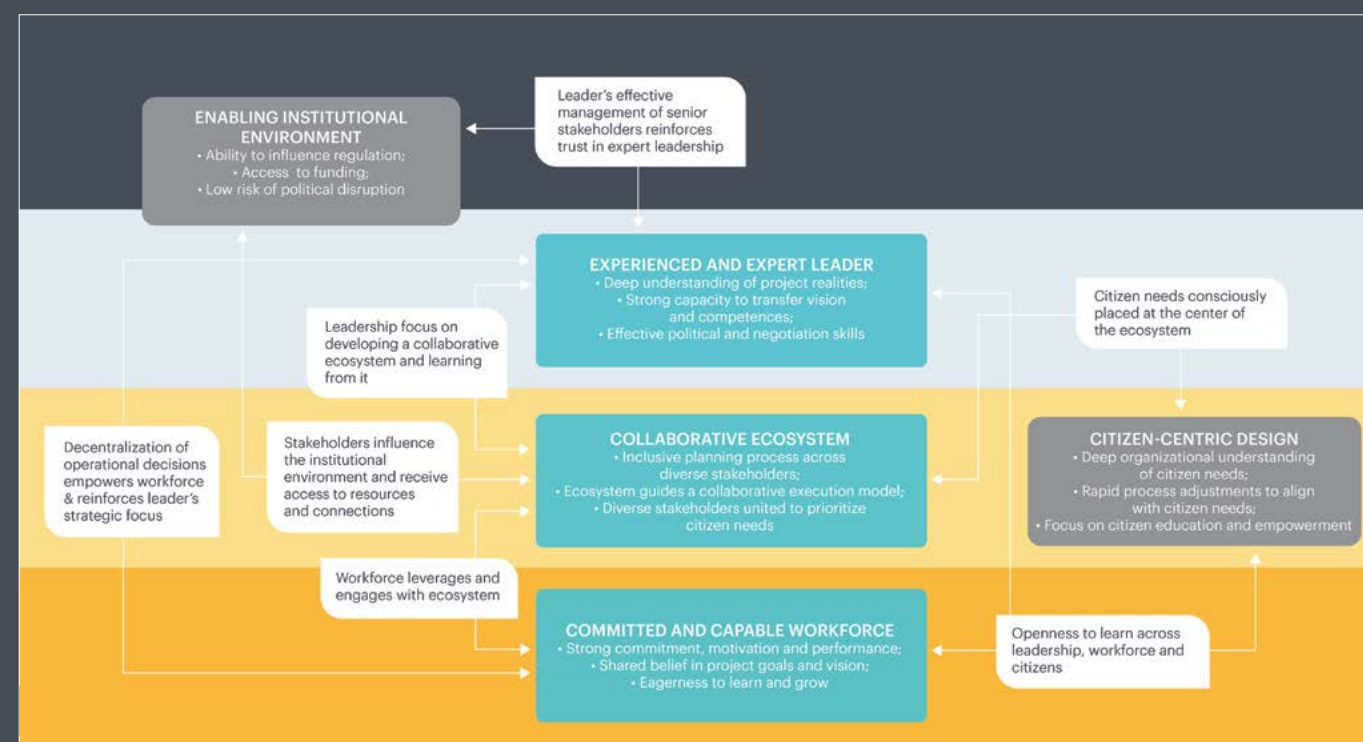


FIGURE 3

The core (blue) and peripheral (grey) drivers of Osasco's land titling success.

- ENVIRONMENT
- LEADERSHIP
- ORGANIZATION
- IMPLEMENTATION
- CORE CONDITIONS
- PERIPHERAL CONDITIONS
- FACILITATING MECHANISMS



CORE DRIVER 1 AN EXPERIENCED AND EXPERT LEADER

The project leader was an expert on land titling issues with a background as head of the Department of Housing and Urban Development. He gained experience when working in land title regularization in São Paulo. His unique contribution was to develop a clear vision and bring that new perspective to this established community. He effectively transferred that vision to the project staff, and along with the competences and skills required to implement it.

The leader also proved adept at the political aspects of the job and at seeking financial funding from the federal government. He focused on communicating and articulating the project's vision to all stakeholders including political sponsors, citizens, and the workforce, understanding that shared commitment and motivation could make up for scarce resources.

A critical aspect of his vision was to mobilize a collaborative ecosystem that could draw on the skills and motivation of diverse members; he provided enough guidance to ensure alignment but allowed enough decentralized autonomy to encourage empowerment and motivation.

CORE DRIVER 2 COLLABORATIVE ECOSYSTEM

The project's decentralized organizational structure was a critical factor in its success. It enabled rapid execution as middle managers and the technical staff were empowered to drive ahead with operational activities. In the meantime, the leader worked on building strategic alliances and engaging with political stakeholders across several governmental levels. The organizational structure created a wide collaborative network. This involved the creation of forums where citizens and other stakeholders could contribute ideas to policies and plans; agreements with real estate registry offices to accelerate the dispatch of documents; and the creation of a local council to unite stakeholders. The network also included the establishment of partnerships with federal government agencies for funding, for example; with the Turin Polytechnic School to develop urban technical plans; and with private firms to gather information and build databases. The result was a highly collaborative and inclusive planning system, centered on citizen needs.

CORE DRIVER 3 COMMITTED AND CAPABLE WORKFORCE

The workforce's high performance resulted in part from the learning and sharing of ideas through the project's ecosystem of partners. The spirit of openness to learn and grow was infectious, and guided by the clarity of the shared vision and values across the workforce. Motivation and commitment were kept high thanks to the autonomy attributed to them by the leader and decentralized model. This virtuous cycle of empowerment, motivation and learning was critical to success, especially as workers were not offered financial incentives nor allotted dedicated time for the project.

PERIPHERAL DRIVER 1 A CITIZEN- CENTERED DESIGN

The participation of the communities targeted by the program was an important factor for success. Citizens recognized that their needs were being placed at the center of the collaborative effort, and that the plans and processes were continually adjusted based on their input. In order to ensure genuine relevance to the citizens, direct interaction was necessary between the workforce and the slum inhabitants, allowing the team to accumulate in-depth knowledge of the target population and address pressing problems as they arose, in a straightforward way. The learning process was intense, but it allowed deep citizen engagement as people saw the system working for them, keeping the ultimate goal of improving their livelihoods front and center.

PERIPHERAL DRIVER 2 AN ENABLING INSTITUTIONAL ENVIRONMENT

The political and institutional environment played an important role in the collaborative ecosystem. For example, the impact of bureaucratic hurdles was limited because the team worked in partnership with institutions to modify the relevant legislature. Likewise, relations with political players were handled smoothly, with effective communication from the project leader being rewarded by trust and support from the political sponsors. In fact, the project leader's adept management of relationships with senior political figures was crucial in accessing resources and connections that enabled the ecosystem to flourish and the virtuous cycles to spin.

REIMAGINING MASS TRANSIT: HOW BRT CURITIBA MOVES PEOPLE

THE GRANDDADDY OF ALL BUS RAPID TRANSIT SYSTEMS SERVES 75% OF CURITIBA’S COMMUTERS, WHO GIVE IT A NEARLY 90% APPROVAL RATING.

Good ideas can be timeless. Just ask the people who rely on one to get around as they live and work in Curitiba, Brazil’s eighth most populous city. Building on the “Agache Plan” from the 1940s and implemented in the 1970s, Curitiba’s bus rapid transit (BRT) system has since evolved into the archetype of - and still-vibrant model for - BRTs worldwide. In 2015, the system served more than 1.5 million passengers per day, via over 1,000 specially-designed buses, and employed about 15,000 people in roles as varied as bus drivers, ticket sellers, mechanics, land integration specialists and city planners.¹⁰

INNOVATING TO INTEGRATE

The BRT’s spoke-like route design converges on Curitiba’s city center “hub,” enabling users to reach their destinations quickly and with a minimum of trouble. What’s more, its current use of special bus-only routes enables the system to bypass traffic congestion, thus allowing it to rival much more expensive subways or light rail alternatives in terms of timeliness. In fact, constructing the BRT cost ten times less than a monorail system, which itself would have cost ten times less than a subway. In hindsight, the choice seems obvious.

Innovations in the system included the use of elevated platforms that permit passengers to walk straight into buses instead of having to climb stairs, similar to a metro; the design of 270-passenger buses, overcoming the usual requirement of having a driver for each 40 to 60 bus passengers; and a ticketing system that allowed a single ticket to cover trips involving multiple buses, another lesson from the metro.

Estimates suggest that 75% of the city’s commuters use the system. What’s more, BRT Curitiba users appear to love it, with one study indicating a nearly 90% approval rating.¹¹

STUMBLING OVER LAND USE ISSUES AND AD HOC SOLUTIONS
A significant challenge in implementing the BRT Curitiba system resulted from a common shortfall among cities in developing economies - the difficulty of integrating land use and public transportation planning. Public transport solutions in Brazil tend to be based on incremental improvements rather than large-scale transformation, the latter being the ambition in Curitiba. This could mean significant upheaval of existing land use. To succeed, BRT Curitiba needed long-term, creative—and potentially disruptive - solutions, and it needed to implement them with scarce financial resources.



HOW THEY DID IT
Three core interacting conditions drove the project’s success: a stable and supportive political regime, an innovative and visionary leader, and a highly empowered workforce. The research also highlighted two peripheral enablers: an openness to work with external partners and strong project planning (Figure 4).

Curitiba, Brazil

TRANSPORT BUS RAPID TRANSIT (BRT)

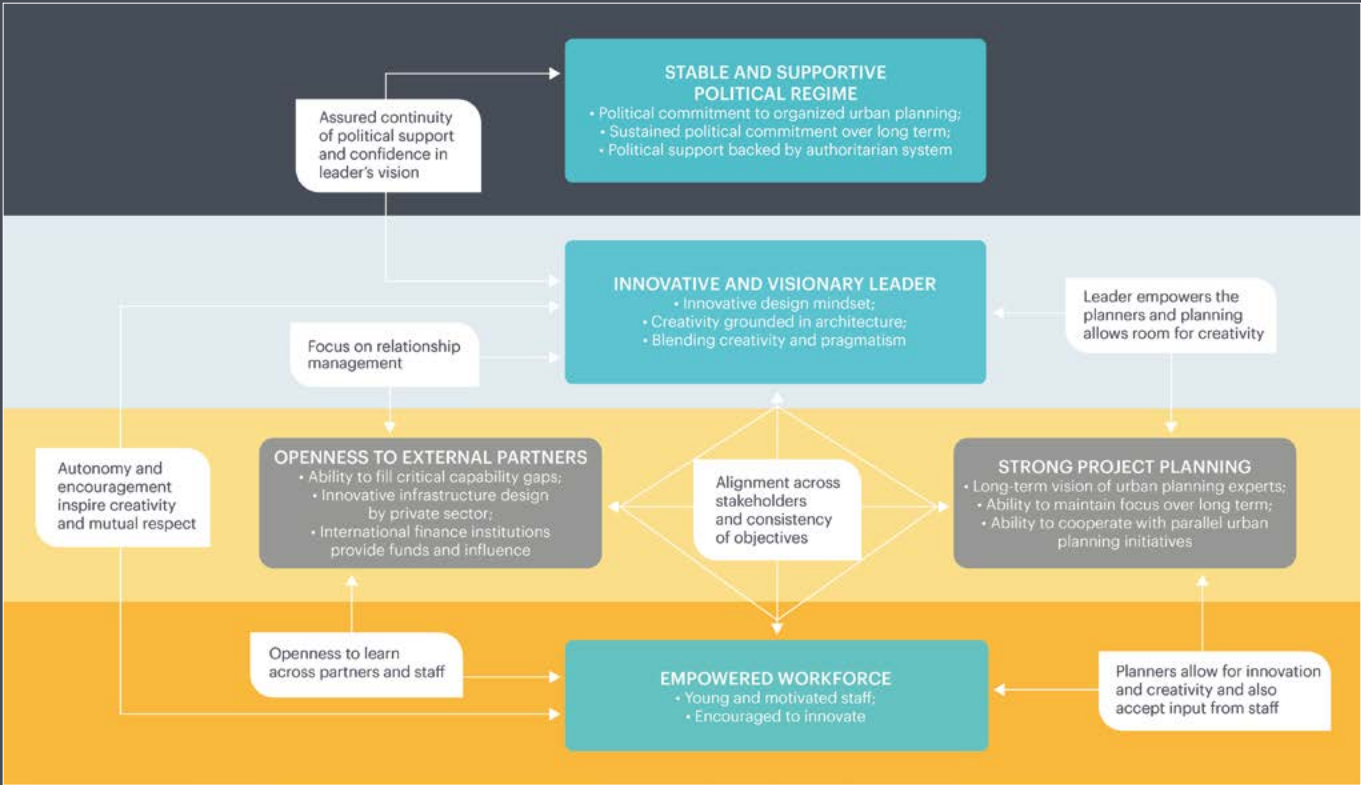


FIGURE 4
The core (blue) and peripheral (grey) drivers of BRT Curitiba’s success.

- ENVIRONMENT
- LEADERSHIP
- ORGANIZATION
- IMPLEMENTATION
- CORE CONDITIONS
- PERIPHERAL CONDITIONS
- FACILITATING MECHANISMS



CORE DRIVER 1 STABLE AND SUPPORTIVE POLITICAL REGIME

Given the complex, long-term nature of the BRT Curitiba project, success required sustained political support, which itself depended on the stability of the regime in control. Curitiba offered this type of long-term political continuity, both in terms of the BRT ecosystem and the vision of the project leader. The project began under Brazil's military regime, which ruled from 1964 to 1985, and continued under the country's reassertion of democracy that immediately followed. In both situations, political support for the BRT remained strong.

CORE DRIVER 2 INNOVATIVE AND VISIONARY LEADER

Jaime Lerner, who served as mayor of Curitiba for three terms - twice during the military regime and a final time during the democratic era - drove the inspiration for the project's current manifestation. A renowned architect and urban planner, he took an active role in supporting and guiding the BRT Curitiba project. Lerner made the critical decision to empower a private company, whose main shareholder was the city of Curitiba, as the operational driver of the project. This allowed the project to benefit from the autonomy, flexibility and innovative strengths of a private firm as well as the citizen-focused planning of the public sector. For example, the public players ensured fair pricing and appropriate routes, while the private status of the firm allowed efficient processes unencumbered by heavy bureaucracy.

Lerner's vision of the BRT took advantage of the city's land use patterns to ensure enough people resided within walking distance of bus stations to make the system viable. Ultimately, the BRT stations themselves attracted more people to live close to them, driving the evolution of Curitiba's urban landscape.

Jaime Lerner's inventiveness and passion were an inspiration to the workforce and indeed, he saw it as an important part of his role to encourage and fuel innovation and new ideas across the organization.

Cássio Taniguchi, Lerner's BRT Curitiba successor and also a former mayor, helped to guarantee the project's continuity by adhering to its original vision. Taniguchi continued the approach of integrating high population areas, the BRT and the city itself in a way that did not degrade Curitiba's historical downtown area. What's more, the design enabled future expansions to handle the city's natural growth without major upheavals.

CORE DRIVER 3 AN EMPOWERED WORKFORCE

The implementation of the project depended on a young, motivated workforce tasked to develop innovative solutions in an environment with limited financial resources. BRT staff contributed to a fluid process of continuous feedback and intervention, which also included other stakeholders. It was this spirit of openness to share and cross-pollinate ideas that stimulated innovations like the larger, longer articulated buses and the elevated platforms. The governing idea was that of a hybrid "train with wheels."

The freedom to generate ideas, the satisfaction of having those ideas taken seriously, and the empowerment felt by seeing those ideas implemented, reinforced the motivation and commitment of the workforce to keep innovating.

PERIPHERAL DRIVER 1 STRONG PROJECT PLANNING

The project's complicated planning and execution required high levels of cooperation and coordination across a diverse group of stakeholders, both inside and outside of government. The two most important participants were the IPPUC, a semiautonomous agency that's part of the city's Urban Planning Institute, and the URBS, Curitiba's urbanization office. While the IPPUC focuses on implementing the master plan and making sure it aligns with the project's master plan, the URBS controls the execution of services, manages fares and takes responsibility for financial performance.

The two agencies worked together closely and were the source of much innovation. For example, a key goal involved preserving the historic section of the city center, which was a challenging area to transport passengers through rapidly and at low cost. The solutions they developed avoided the need for traditional bus terminals by designing innovative bus stops—essentially small-scale stations more similar to the design of metro systems—that allowed simple connections between bus routes. This was more appropriate for the cramped city center and allowed faster boarding processes. URBS also worked in partnership with Volvo to design the "bi-articulated" buses with drivers stationed at both ends, so they did not have to turn around in the center's narrow historic streets.

PERIPHERAL DRIVER 2 OPENNESS TO EXTERNAL PARTNERS

Interacting with the private sector enabled the project's public sector workforce to innovate more expansively. Examples of innovations that emerged from these public-private interactions included the construction of bicycle lanes along the BRT paths, the use of native vegetation in the landscape, biodiesel-powered buses and even the design of the futuristic tubular bus stations. The project's collaboration with Volvo was central in developing the famous articulated buses. The private communications firms Vivo telecom and Ericsson collaborated to improve fleet management and network planning. Their input led to smarter ticketing systems, reliable transport information, and better wireless connectivity for passengers. These partnerships resulted in tangible improvements such as reduced passenger complaints and programs to reduce CO2 emissions.

International organizations such as the Inter-American Development Bank and the Agence Française de Développement also partnered with the municipality through the provision of low-interest loans and technical support, mostly focused on sustainable environmental development and social integration.

A CURE FOR BUREAUCRATIC INERTIA: SAVING TIME IN SÃO PAULO

HOW BRAZIL'S LARGEST CITY TRANSFORMED ITS CUMBERSOME PUBLIC BUREAUCRACY INTO A LEAN, CITIZEN-FRIENDLY MACHINE.

The largest city in the Southern Hemisphere, São Paulo is home to nearly 12 million people who, from time to time, need to renew or replace their driver's licenses or ID cards. And when they do, the creaky wheels of the traditional public bureaucracy used to slow things down to a crawl—so much so, in fact, that a simple act like a driver's license renewal could take months to complete fully. Why so long? Most Brazilians blamed the “despachante”—private intermediaries who supposedly facilitated the process via their connections in the bureaucracy, but often resulted in slower progress and increased costs. Other impediments included the geographically distributed nature of the tasks required to fulfill an application. For example, to replace a lost or stolen identity card, the citizen had to provide a new fingerprint sample and fill out an application at an agency, which then shipped both to a central office where staff compared the new prints to those already on file. If they matched, the citizen would receive a replacement card. The whole process could take up to 60 days to complete. As public outrage over the situation increased, government leaders searched for the most effective way to speed things up. Here's what they did.

ONE-STOP-SHOPPING WITH “POUPATEMPO”

Their solution, “Poupatempo” (Portuguese for “time saver”), debuted in 1997 and drew inspiration from other successful experiments, including a similar transformation in the Brazilian State of Bahia. They introduced a “one-stop-shopping” model that draws together all the different processes involved in license renewal, for example, into one place, thus effectively eliminating the shady intermediary channel. The change cut the driver's license renewal process to under 77 minutes, delivering an 80% to 85% efficiency boost compared to the traditional method. Researchers put the cost savings at R\$ 96 (roughly US\$ 30) per person. About 3 million citizens use the service each year for total time savings of 162 million minutes, which over five years saves about R\$53 million (roughly US\$ 16.5 million).

SUCCESS BREEDS SUCCESS

In 2016, 72 Poupatempo locations provided over 1000 services across São Paulo, including the issuance of national driver's licenses, identity cards, criminal record certificates and vehicle licensing¹². The program is so successful that it has become a model for other countries that are wrestling with the same problems.

In 2017, Poupatempo expects to reach 50% of services in virtual channels and to add new services related to Health and Public Security. Further innovations in the pipeline include integration with an international fingerprint system and partnership with Brazil's leading banks to allow electronic payments through ATMs around the city¹³.

HURDLES: CO-LOCATION, INTERMEDIARIES AND MAKING “FAST” MORE EFFICIENT

The Poupatempo team faced three significant hurdles in creating the new approach:

1. Physically co-locating formerly separate public services. For example, renewing a driver's license requires input from agencies such as São Paulo state's Department of Traffic (DETRAN) as well as municipal and state police agencies, all of which are in different locations.
2. Dealing with the intermediaries. Implementing Poupatempo effectively was always going to be a challenge due to the political influence of the intermediaries.
3. Faster doesn't always mean more efficient or better. Poupatempo's focus on time could cause government employees to cut corners. For example, driver's license applicants need to pass a physical examination, which in a one-stop-shop setting could become rushed and less accurate compared to the traditional process.

HOW THEY DID IT

Poupatempo's successful transformation was achieved through three core interacting conditions (Figure 5): strong political support, strong leadership commitment, and a highly motivated workforce. The research also pointed to two peripheral influences that reinforced the successful dynamic, namely effective collaboration across government agencies and strong citizen demand for change.

São Paulo, Brazil

PUBLIC BUREAUCRACY POUPATEMPO

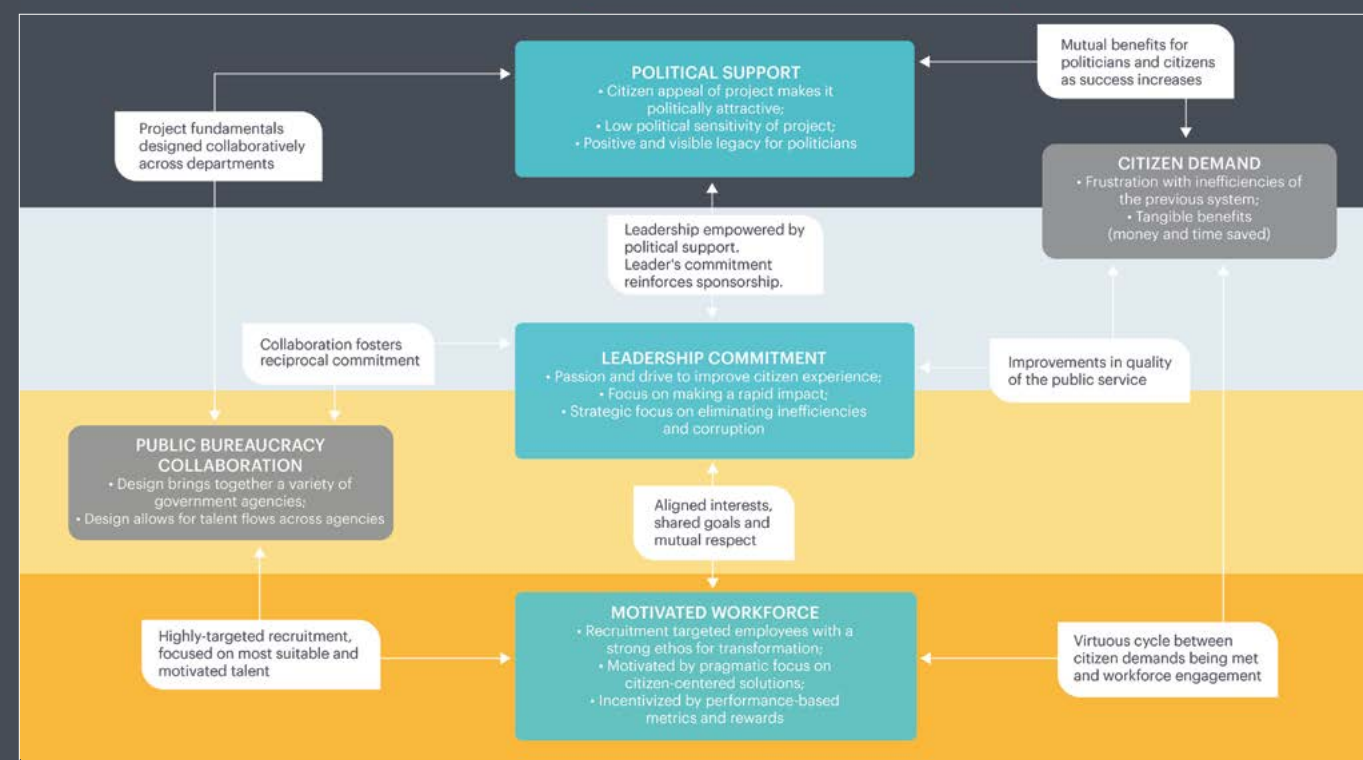


FIGURE 5

The core (blue) and peripheral (grey) drivers of Poupatempo's success.

- ENVIRONMENT
- LEADERSHIP
- ORGANIZATION
- IMPLEMENTATION
- CORE CONDITIONS
- PERIPHERAL CONDITIONS
- FACILITATING MECHANISMS



CORE DRIVER 1

STRONG POLITICAL SUPPORT

Strong political support was invaluable to the Poupatempo project, and the enthusiasm of citizens for the changes made this an easy decision for politicians and bureaucrats. This set up a virtuous cycle between politicians and the citizenry as success bred further enthusiasm and support for the political sponsors of the project.

The political support helped the project overcome a number of formal restrictions that otherwise would have proved intractable. For example, processing some documents required access to often heavily restricted police records. Poupatempo's strong political clout made it possible for the city's executive branch to place authorized police staff in the one-stop building to help with processing.

Likewise, while intermediaries stay in business by cultivating strong political ties, the citizen interest in seeing Poupatempo succeed overwhelmed their efforts to stop it. Strong voter demand made any move to impede the program politically risky. The stable political environment in São Paulo during the program also played a role, with the same political party in power at the state level throughout the period. Of course, political support also meant easier access to the substantial financial, physical and human resources the Poupatempo project needed to access.

The confidence of the political sponsors increased because of their trust in the project's highly effective and focused leadership and it was reinforced by leader's ability to deliver results.

CORE DRIVER 2

STRONG LEADERSHIP COMMITMENT

Our interviews uniformly identified the leader of the Poupatempo project, São Paulo's then-governor, as the individual who played the pivotal role in nurturing and executing the initiative. Taking cues from similar projects in other countries and Brazilian states, he defined the project's goals and indirectly supervised Poupatempo's implementation. The project's high political visibility and strong leadership enhanced its importance, and led to the establishment of a strong technical advisory board tasked to make Poupatempo happen. Likewise, the governor's strong emphasis on eliminating inefficiencies, such as the intermediaries, in order to fulfil citizen needs helped to reinforce the project team's commitment to the undertaking.

For the project to succeed, the project lead had to be respected and trusted not only by his workforce but also by citizens, by his political sponsors and by stakeholders across government agencies with whom his model demanded collaboration. A key part of his role, therefore, was to ensure this wide array of stakeholders shared the same vision and motivation to achieve their shared goals.

CORE DRIVER 3

A HIGHLY MOTIVATED WORKFORCE

Project leaders chose staff based on technical criteria, and something more. Most important, they were selected from a variety of public service backgrounds and united by the fact that they expressed a passion to radically disrupt and improve the way in which citizens receive public services. They wanted to be part of a movement for progress and change in public service delivery. So, as the project successfully implemented the innovative model and gauged the positive citizen reaction, the workforce and citizens reinforced one another's enthusiasm and satisfaction levels.

In addition to enhancing their career choices, job candidates received both monetary and non-monetary incentives, with the latter including the opportunity to make a positive impact on society - something they prized highly. Those selected for recruitment participated in training and feedback sessions to improve service quality. And to sustain these service levels, the project's management committee continually tracks a variety of key performance indicators such as processing times and customer satisfaction levels.

PERIPHERAL DRIVER 1

STRONG COLLABORATION ACROSS GOVERNMENT AGENCIES

Our interviews revealed the high levels of collaboration that took place among a wide variety of participating government agencies, including those focused on security, transport and work and employment. From the project's initial meetings, the leaders of these and other agencies worked with the governor to define the project's goals and how they would meet them. When disagreements did arise, the governor often provided the energy and influence required to resolve them. This collaboration extended to staffing decisions, enabling the project to secure the people it wanted from different government agencies.

PERIPHERAL DRIVER 2

STRONG CITIZEN DEMAND

A significant source of power and momentum that animated Poupatempo's success flowed from citizen dissatisfaction with the status quo. People were tired of the endless waiting times, having to deal with the intermediaries and the costs involved—all of which the project quickly and effectively addressed. This tidal wave of public opinion cascaded through São Paulo's political and bureaucratic institutions, ultimately proving unstoppable and powering the continued evolution and expansion of the project. Interacting with positive and motivated staff in the new system only reinforced the citizens' conviction that this new service was worthy of the public, and vice versa.



READ ALL ABOUT IT: PAYING TEACHERS TO IMPROVE

SCHOOLS IN INDIA'S STATE OF ANDHRA PRADESH DISCOVERED A WAY TO DRAMATICALLY REDUCE ILLITERACY. THE SECRET? AN INCENTIVES SYSTEM ADAPTED TO ITS CONTEXT.

Illiteracy is a major problem across India, where the 2001 census indicated that fewer than 65% of the population could read.¹⁴ In the country's seventh largest state, Andhra Pradesh, rates were even lower at about 60%.¹⁵ To address the situation, administrators chose five of the state's then-23 districts to pilot a teacher pay-for-performance initiative.

FINDING CROWDED CLASSROOMS AND SINGLE-TEACHER SCHOOLS

The five chosen districts lacked sufficient education faculty and suffered from crowded classrooms. The districts also had large numbers of single-teacher schools—in one district, Nizamabad, over 60% of schools had only one teacher. Single-teacher schools make it practically impossible to give individual attention to students, increasing workloads and having a negative impact on learning.

The teacher pay-for-performance solution launched in Andhra Pradesh reflected work done by the World Bank, scholars from Harvard University and the Azim Premji Foundation. These experts partnered with the state government to evaluate different options for improving the quality of primary education. Specifically, the goal of the program was to learn whether performance-based teacher payments could produce results. To find out, the program selected 500 schools, and grouped them according to the type of benefit they would receive. The two options were providing schools with additional inputs such as an extra contract teacher or a cash grant, or offering teachers cash incentives for achieving better performance via group or individual bonuses based on student performance.

MEASURING REAL IMPACT

Originally slated to run for six months, the program ultimately endured for five years due to its success. Administrators noted the impact it not only had on students, but also on teachers and on the overall education performance of involved districts.

Within two years of the program's launch, clear differences emerged in the performance of schools with incentivized teachers compared to the schools without such incentives.

What's more, schools that received extra contract teachers and cash grants performed better than those without such incentives. The program also generated spill-over effects in other subjects such as science and social studies, where student performance improved despite the exclusion of these courses from the program. Crucially, the way the incentives were designed and implemented generated motivation among the teachers and schools, and this was reinforced by the chance to actively influence the project's design. As positive results became visible, so did the participants' enthusiasm, motivation and commitment to the new model. In quantified terms, the teacher performance pay initiative increased rural literacy by 20%, boosted teacher motivation levels by 75%, and compelled 76% of teachers to make special efforts with students. Likewise, 85% of teachers favored the bonus payment approach to improve student performance.¹⁶

At the same time, the school districts drove improvements in student-to-teacher ratios and reductions in the number of single teacher schools across the five districts. Compared to similar programs in other countries, India's performance pay experiments in Andhra Pradesh did better in several areas, such as the strength of the program's core design, overall incentive levels and observed effects.¹⁷

HOW THEY DID IT

Andhra Pradesh's teacher performance incentive program worked through the interaction of three core elements: Skilled project leadership, an innovative program design, and a robust collaborative model. It also benefited from two peripheral reinforcing conditions: Stable administrative and financial support and motivated teachers and schools (Figure 6).

East Godavri, Vishakapatnam, Kadapa, Medak, Nizamabad - India

EDUCATION TEACHER PERFORMANCE PAY INITIATIVE

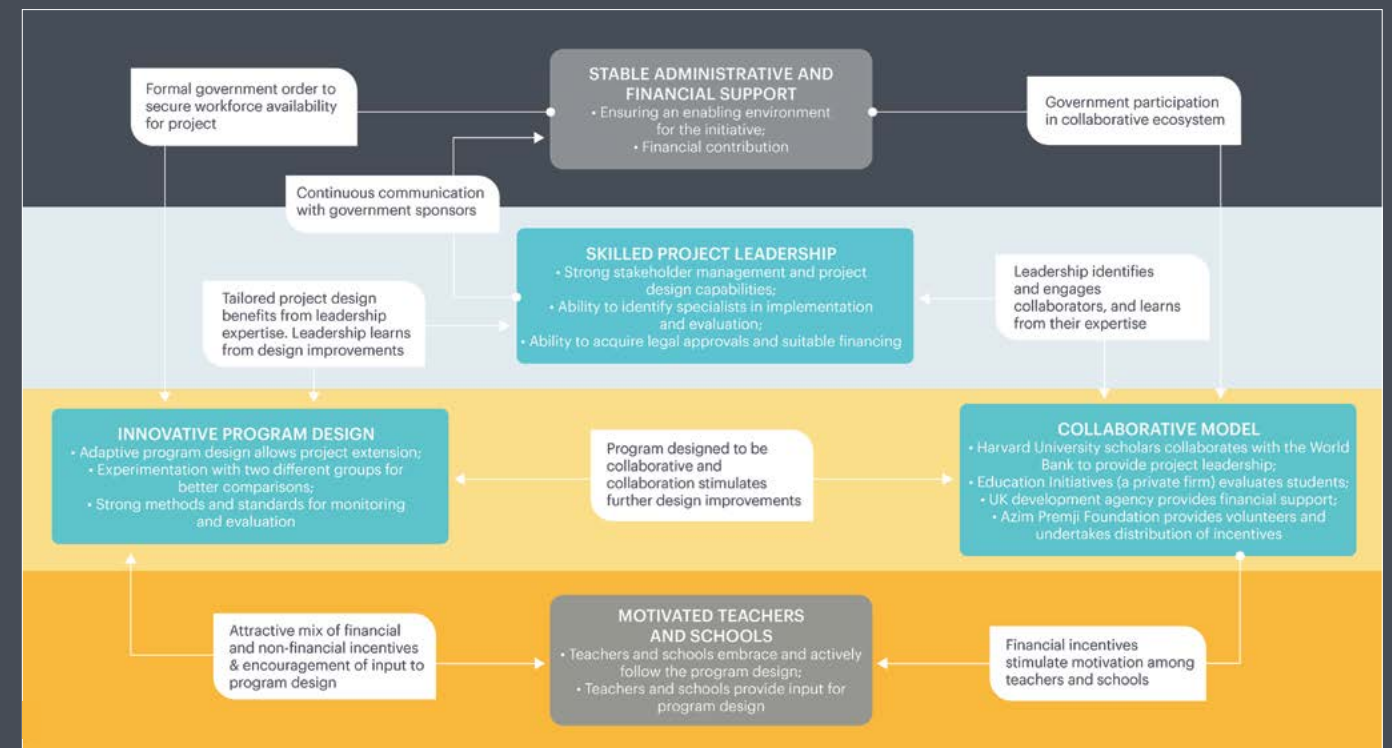


FIGURE 6

The core (blue) and peripheral (grey) drivers of the Andhra Pradesh Teacher Performance Pay initiative's success.

- ENVIRONMENT
- LEADERSHIP
- ORGANIZATION
- IMPLEMENTATION
- CORE CONDITIONS
- PERIPHERAL CONDITIONS
- FACILITATING MECHANISMS



CORE DRIVER 1
**INNOVATIVE
PROGRAM
DESIGN**

One of the first randomized evaluation experiments in India, project leaders designed the program with the flexibility that the situation demanded. For instance, after the first round of interventions where teachers received incentives based solely on improvements in the performance of students, the second iteration introduced more variables and asked teachers to hit specific targets for improvements to be eligible for incentives. Having a coalition of experts involved and a collaborative, flexible model, ensured that the program evolved to better fit its context and deliver higher performance. The ability to experiment with two different groups—the first with direct teacher incentives, the second focused on other forms of support - enabled the project leaders to make more accurate comparisons and tailor the design accordingly.

The project also relied on effective methods and standards for monitoring and evaluating results. Leaders introduced systems to detect corrupt practices, measure volunteer performance and ensure the accurate collection of data for final evaluations. For instance, they introduced monitoring mechanisms to track when the volunteers were uploading data, who was uploading it, whether there were any delays, and why. The code input process could detect wrong data entries, alerting the authorities immediately regarding who had logged them in and when.

CORE DRIVER 2
**SKILLED
PROJECT
LEADERSHIP**

The team that lead the Andhra Pradesh Teacher Performance Pay initiative included experts from Harvard University and the World Bank. These experts were involved in designing a comprehensive program, identifying and bringing on board specialized organizations and arranging for adequate financial resources.

The leadership team identified suitable sources of financing such as the Department for International Development (DFID) under the UK Government and specialists such as the Azim Premji Foundation and Education Initiatives for the implementation and evaluation of the experiments. The team also acquired the necessary legal approvals for carrying out the experiments in the country and sold the program to the state government in India.

CORE DRIVER 3
**COLLABORATIVE
MODEL**

A collaborative ecosystem played an instrumental role in the success of the program, bringing together all stakeholders responsible for its execution. At the core were the experts from Harvard and the World Bank, who not only designed the flexible program but also identified specialists for its implementation and evaluation.

For example, the Azim Premji Foundation, a not-for profit organization that has been working for more than a decade to improve the quality of education in India, took a leading position in the program's operational execution. The foundation played two important roles: first, it brought in the volunteers who implemented and monitored the program and second, it routed the payments to the teachers and schools. Likewise, Education Initiatives, India's leading psychometric testing company, designed and administered the tests to evaluate student performance improvements.

Finally, these partners also provided technological resources that enabled the program's smooth implementation. Field workers were equipped with laptops and tablets to collect information such as student raw test data that the program then evaluated. These and other technologies also enabled effective performance monitoring.

PERIPHERAL
DRIVER 1
**STABLE
ADMINISTRATIVE
AND FINANCIAL
SUPPORT**

While the political leadership did not involve itself in the implementation of the experiments, it was instrumental in getting the experiments up and running. The support of government officers such as the Education Secretary played an important role here. In fact, the Education Secretary clearly understood the goals of the program and thus the government agreed not to transfer teachers within the state while the program was underway. The monitoring, feedback loops and focus on communication were critical in keeping the government informed, maintaining their support and minimizing interference.

Financial resources were particularly important for the program since they enabled leaders to hire specialized organizations such as the Azim Premji Foundation and Education Initiatives, creating an attractive incentive pool for the participants and allowing flexibility in program design. Again, professional project management, monitoring, clear feedback and communication were important mechanisms to build the confidence and trust among the sponsors.

PERIPHERAL
DRIVER 2
**MOTIVATED
TEACHERS AND
SCHOOLS**

The teachers and schools involved in the program exhibited high levels of interest in participating in the initiative. They embraced and actively engaged in its design and implementation. Collectively, they provided critical design input, including teaching and coaching techniques and methodologies. They also developed the role of financial incentives within the model. Being able to contribute actively to the project design was an important way to build the commitment and motivation of the participants. This reinforced the motivating effect of performance-linked financial and non-financial incentives.



SUPPLYING WATER FOR A THIRSTY HYDERABAD

WHILE INDIA'S HYDERABAD HAS EXPERIENCED TREMENDOUS DIGITALLY-DRIVEN GROWTH, ITS WATER SYSTEM REMAINED DECIDEDLY LOW TECH UNTIL AN INNOVATIVE PROGRAM QUENCHED THE CITY'S RAGING THIRST.

A major Indian technology hub, Hyderabad faced a serious water shortage in the 1980s as the city's population growth began to overwhelm its decrepit water and sewerage networks. Between 1965 and 1991, Hyderabad metro's population grew 330%, from 1.3 million to 4.3 million. Over the same period, the per-capita daily water supply nearly halved, from 192 liters to 106 liters.¹⁸ The pressure on water supply combined with a fragile sewerage system resulted in common outbreaks of diseases, such as cholera.

Complicating the situation, the two municipal agencies that controlled the water and sewerage services - the Public Health Engineering Department and the Municipal Corporation of Hyderabad - faced declining revenues and increasing operational costs.

CREATING A SEMI-AUTONOMOUS SOLUTION

The solution Hyderabad came up with in 1989 consolidated the two agencies into a single semi-autonomous body - the Hyderabad Metropolitan Water Supply and Sewerage Board (HMWSSB). To increase transparency, ensure better services to customers and build a more accountable organization, the HMWSSB launched four key initiatives in the late 1990s:¹⁹

THE CUSTOMER MEETS CAMPAIGN

An opportunity for customers to engage in face-to-face interaction with senior officials. During these meetings, citizens highlight issues they're having related to the water supply, sewerage service and billing procedures.

METRO CUSTOMER CARE

A 24/7 computerized helpline to receive and monitor complaints and coordinate responses.

SINGLE WINDOW CELL

A dedicated office to receive, process, and coordinate all new water and sewerage connection applications.

METRO WATER CITIZEN'S CHARTER

A set of service delivery standards and accessible grievance redressal procedures that earned substantial goodwill from the public and consumer rights groups.

These four initiatives initially increased HMWSSB's revenue collection by 24% in 2000, by a further 18% in 2001, and another 36% in 2002. Furthermore, nearly 280,000 customers who had previous outstanding bills received an opportunity to make a one-time settlement. As a result, monthly collections improved from a maximum of US\$ 1.3 million (INR 60 million) to an all-time high of US\$ 3.98 million (INR 184 million) in 2004. Overall, more than 17,000 chronic defaulters paid up their debts.²⁰

While originally initiated in 1999, most of the reforms continue to this day and have undergone several iterations to improve service quality for citizens. This underlines the dynamism of the model, driven by monitoring, transparency and continual improvement. Compared to the water supply and sewerage board in Bangalore, for example, the HMWSSB provides better water coverage and higher daily water production per capital; experiences far fewer pipe breakage incidents; and spends nearly three times less for each 100 cubic meters of water.²¹

TOILING TO IMPROVE TRANSPARENCY

Achieving transparency and accountability was a challenge as it required a major transformation of entrenched cultures and behaviors within the workforce. The previous system encouraged employees to complicate customer interactions with the aim of generating additional private income and bribes, for example, by channeling customers toward plumbers outside the official public system to solve their issues. A new, transparent system demanded an entirely new culture and attitude among workers and their interaction with customers.

HOW THEY DID IT

The Hyderabad project succeeded thanks to the interaction across four core drivers and a peripheral one. Committed political sponsorship, high-caliber leadership, transparent organization and accountable workforce were the core drivers, and collaborative implementation was the peripheral driver (Figure 7).

Hyderabad, India

URBAN PLANNING WATER SUPPLY AND SEWERAGE BOARD

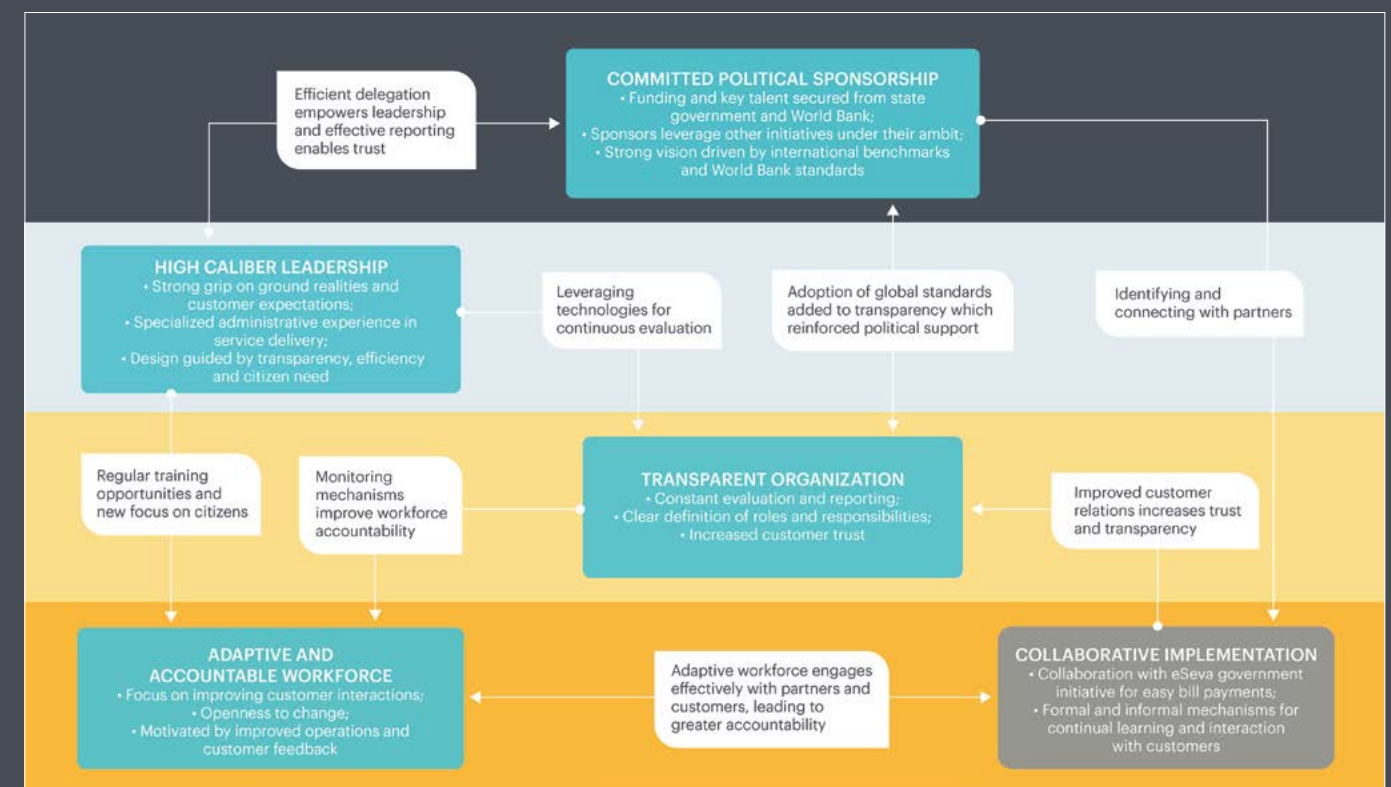
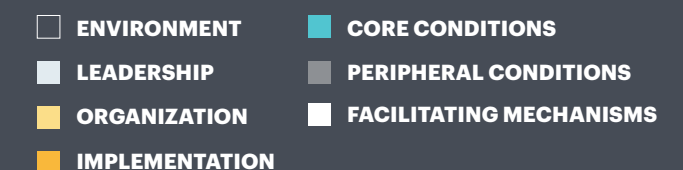


FIGURE 7

The core (blue) and peripheral (grey) drivers of success behind the Hyderabad Water Supply and Sewerage Board project.





CORE DRIVER 1 COMMITTED POLITICAL SPONSORSHIP

The chief minister at the time envisioned the HMWSSB as a public utility that would deliver better services while also achieving financial self-sufficiency. Both he and his successor agreed on the need to reorganize the existing structures to create the new HMWSSB organization.

As the key political sponsor, the chief minister was extraordinarily proactive in his participation. He was designated as Chairman of the Board at HMWSSB, ensuring close strategic alignment between the project and all its sponsors. This role facilitated the securing of funding and key talent from the State government and from the World Bank. It also made it easier for the project to connect with and make use of other initiatives in the state to support the growth of HMWSSB.

The political sponsors, led by the chief minister, were also crucial in insisting upon transparency at the core of the project. The project realized this vision through the application of international benchmarks and World Bank standards. The establishment of those global standards reinforced trust and confidence in the transparency of the system, and thereby secured further political support.

A key decision by the political sponsors was to select a highly-skilled civil servant to lead the complex project—one specialized in administrative change. The sponsors empowered the project leader through effective delegation, and in return, his clear reporting reinforced their relationship of trust.

CORE DRIVER 2 HIGH-CALIBER LEADERSHIP

The top management team was very skilled in administration and service delivery, crucial capabilities for this project. They had a keen understanding of how to use technology to conduct continuous evaluations, ensuring transparency and clear reporting of progress. The effective reporting gave sponsors confidence and maintained their trust and support.

Service delivery in this case meant understanding the reality of citizen needs and developing an organization that responded to those needs. The leader transferred this new citizen-centric approach across the workforce, investing in training and a variety of initiatives to support this objective. For instance, as a part of the Customer Meets Campaign, managers set up “open darbars” (literally, “open talk”) to collect feedback from stakeholders and encourage citizens to share their ideas. Ultimately, this dialogue enhanced the accountability and authority of the organization and the trust thus gained resulted in significant cost recovery as citizens began paying their overdue bills

CORE DRIVER 3 A TRANSPARENT ORGANIZATION

Organizational transparency enabled the project to regulate the revenues of HMWSSB and clearly define the roles and responsibilities of the employees. The initiative constantly monitored service level standards for each geographic area of the city to increase accountability among the frontline staff. Additionally, the project evaluated the employees regularly based on service standards, especially regarding their ability to address citizen grievances. The system consistently and carefully used key performance indicators and supporting metrics to monitor progress, improving both customer service and operations. The project introduced technology to allow this continuous evaluation and effective reporting.

From the citizen’s perspective, initiatives like the Metro Water Citizen’s Charter and the 24/7 helpline supported the sense that this really was a service for the public, taking their input and feedback into account. The knowledge that the project was designed to meet international standards and guidelines also reinforced trust in the initiative

CORE DRIVER 4 AN ADAPTIVE AND ACCOUNTABLE WORKFORCE

The chief minister systematically hired top level talent for the HMWSSB. The project recruited well-regarded state civil servants in top management positions in addition to consultants from NGOs and private organizations. The leadership team deliberately focused on providing soft skills such as training programs to help employees better handle interactions with low income communities and to instill an openness to change in employees. This resulted in a genuine desire among the workers to connect better with the real needs of their customers, the citizens.

One example was the HMWSSB’s “Little Champions” effort, which involved employees from all levels actively engaging with citizens to create awareness about the project, its purpose and its implications. Moreover, as workers saw the improvements in their operations and felt the positive impact it was having on citizens, their motivation grew, setting off a virtuous cycle of effort and reward. Monitoring and feedback processes were crucial to making these improvements tangible, but none of it would have happened if the workforce were not culturally open to adapt and grow in this way.

PERIPHERAL DRIVER 1 COLLABORATIVE IMPLEMENTATION

The HMWSSB actively collaborated with other government organizations. For instance, HMWSSB tied up with eSeva kiosks in the city to improve billing collections for the water board. The digitally-enabled eSeva kiosks provide a range of e-governance services to the public, including the ability to pay utility bills. As of 2010, citizens paid over 80 percent of HMWSSB bills at eSeva kiosks across the state. The chief minister played an important role in making these kinds of connections possible across potential partners and collaborators outside the project.

An important part of the continuous, collaborative improvement process was the interaction between the workforce and its customers. Initiatives such as the Customer Meets Campaign, the Metro Customer Care, Single Window Cell and Metro Water Citizen’s Charter, allowed for continuous engagement and input from citizens. The project consequently evolved in a way that became increasingly relevant and responsive to the realities of Hyderabad’s citizens.

TICKET TO RIDE: DELHI BUILDS A SUCCESSFUL (AND PROFITABLE) METRO

DELHI COMPLETED ITS METRO SYSTEM AHEAD OF TIME AND ON BUDGET THROUGH A COMBINATION OF STRONG LEADERSHIP, COLLABORATION, ORGANIZATIONAL OPENNESS AND WORKER SKILL.

At the end of the twentieth century, Delhi found itself roadblocked because it lacked a modern public transportation system. With rising disposable incomes, falling vehicle prices, and road travel as the dominant form of transport, the number of vehicles on the city's roads spiked by nearly 700% between 1980 and 2002. During this period the urban population increased by about 240%. Delhi's increasingly overcrowded bus system helped to accelerate car sales as more commuters opted to drive instead of taking public transportation. The number of cars on the road ballooned from 36 per kilometer (km) in the early 1980s to 125 per km by 2002, putting substantial pressure on the road system.²²

CONTEMPLATING THREE CHOICES

To address the increasing levels of gridlock the city faced, Delhi's government considered three transportation options: a Metro system, a light rail transit solution and another approach that would basically maintain the status quo. The government ultimately gave the go-ahead to the Delhi Metro, beginning construction in 1998 and launching operations in a quick four years.²³

By 2015, the Delhi Metro featured six lines with a combined length of 213 km, and 160 stations.²⁴ That same year, approximately 15 percent of the city's citizens used the service – that's roughly 2.5 million passengers every day.²⁵ The system is one of only a handful of public transportation systems worldwide that has operated in the black from its inception and about 30% of its revenue comes from associated sources such as property development and Metro advertising.²⁶

Studies of the system's performance find that the Metro has taken approximately 390,000 vehicles off city roads, significantly cutting both local pollution and fuel expenditure. By way of comparison, Delhi's Metro outperforms a similar system in Kolkata in several important ways. For example, analysis found Delhi Metro's cost per km to be US\$27 million compared to US\$44 million in Kolkata; and the portion it contributed toward the city's total transport levels was twice as high as in Kolkata.²⁷

CHALLENGES ATTRACTING RIDERS AND MAKING LAST MILE CONNECTIONS

While ultimately successful, Delhi's Metro project faced the challenge of convincing a population with no experience using a Metro system to adapt to an entirely new form of transportation. This included ticketing and embarkation systems and procedures, all in an unfamiliar environment.

Another challenge was the Metro's ability to provide "last mile" connections for commuters. The system's planners struggled to design approaches that would extend the reach of the system without deteriorating service quality.

HOW THEY DID IT

The Delhi Metro initiative succeeded through the interaction of three core drivers and three peripheral ones (Figure 8). The core elements were a capable and empowered leader, an autonomous organization, and a collaborative implementation style. Peripheral drivers included committed political support, an adaptive operating model, and a motivated and skilled workforce. Facilitating mechanisms included leadership encouraging an open and flat organizational culture, encouragement of free flow of ideas for improvement, systems and processes encouraging ideas from workforce, people and culture able to adapt, adaptive operations allowing cross sectoral and cross border collaborations, access to top skills and training opportunities, leadership providing training opportunities and job rotation, and leadership nurturing collaborative relationships while the leader's credibility attracts further partners.

Delhi, India

TRANSPORT DELHI METRO

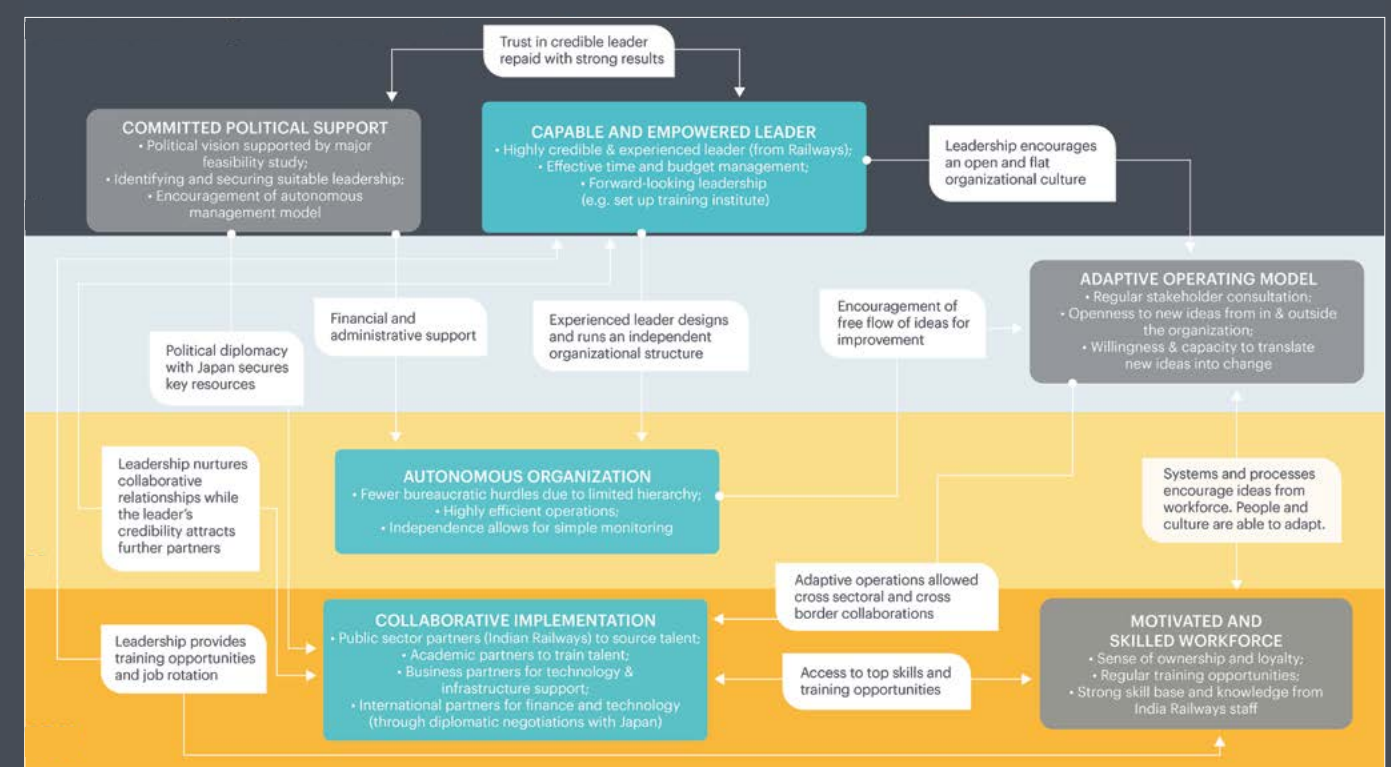


FIGURE 8

The core (blue) and peripheral (grey) drivers of success behind the Delhi Metro initiative.

- ENVIRONMENT
- LEADERSHIP
- ORGANIZATION
- IMPLEMENTATION
- CORE CONDITIONS
- PERIPHERAL CONDITIONS
- FACILITATING MECHANISMS



CORE DRIVER 1 **A CAPABLE AND EMPOWERED LEADER**

The Delhi Metro's first managing director had significant experience with both railways and complex projects. He also enjoyed the complete support of the project's sponsors regarding decisions, which greatly facilitated timely bureaucratic approvals - a key reason for the Metro's completion on time. This focus on timeliness was reinforced by the project leader's adoption of Japanese technologies, cultural practices and energy-saving approaches, which he adapted to the Delhi environment.

The leader designed the project to be collaborative in its implementation, setting up an autonomous organization to operate in an open and non-hierarchical manner, bringing together a variety of stakeholders. The leader nurtured these collaborative relationships and his credibility attracted yet further partners that were willing to participate in this ambitious undertaking.

The leader's decision to implement a non-hierarchical and adaptive organizational model was rare in India's top-down management environment. He established his reputation as a forward-thinking leader, setting up a training institute and ensuring that employees had opportunities to improve themselves as well as the chance to contribute their ideas to improve the project. This approach helped to instill motivation and loyalty.

CORE DRIVER 2 **HIGHLY AUTONOMOUS ORGANIZATION**

Conceived as an independent company co-owned by the Indian and Delhi governments, the Metro's unconventional model enabled its leaders to establish an innovative organizational structure.

This included the delegation of powers, robust transparency and accountability, and a unique work culture, all under the guidance of the managing director. This management approach contrasts starkly with the way traditional government agencies manage mass transit systems across the country. The focus on transparency and accountability led to fewer bureaucratic hurdles, and greatly enhanced the ability to monitor progress. This resulted in more proactive decision-making and financial autonomy, and it ultimately helped the system achieve operational profitability faster.

CORE DRIVER 3 **COLLABORATIVE IMPLEMENTATION**

The Delhi Metro's implementation was achieved through a collaborative ecosystem that provided access to the needed technologies, skills and funding. The project sourced technology from a variety of companies, such as Bharat Earth Movers Limited (India), Bombardier (Canada) and Hyundai Rotem (South Korea). These consortia gave the Metro access to leading global capabilities, enabling it to build world-class infrastructure. The Japan International Cooperation Agency (JICA) also contributed in areas such as construction, electrical and mechanical engineering, and implementation.

Facilitated by the project's sponsors, the JICA also helped the Delhi Metro meet its funding requirements. This financing was extremely important since it came at a time when other public service projects in the country were struggling to access adequate financing. Taking lessons from the Delhi Metro funding success, other Indian cities are now also collaborating with JICA to pursue modernization plans.

There was also collaboration with other parts of the public sector. India Railways proved to be a critical source of talent and knowledge, and academic partners provided valuable training opportunities to workers.

PERIPHERAL DRIVER 1 **COMMITTED POLITICAL SUPPORT**

The Delhi Metro project received strong support from its two co-owners, the city and central governments. It also benefited from the comprehensive feasibility study the two government bodies commissioned on all aspects of the project prior to its launch, including technical, financial and operational issues. Consequently, few surprises occurred during the buildout.

The political leadership sanctioned the creation of Delhi Metro's independent organizational structure and identified able top management to lead the initiative. The sponsors also helped push the project through the inevitable bureaucracy, enabling faster decision making.

PERIPHERAL DRIVER 2 **ADAPTIVE OPERATING MODEL**

The Delhi Metro's independent organizational structure allowed for the free flow of ideas from individual employees, citizens and other collaborating organizations. The operating model expressly allowed and encouraged these exchanges. These interactions and consultations led to tangible new project innovations. For example, the project planners acted on feedback about the absence of toilet facilities at the stations by partnering with Sulabh International to design and construct them. Furthermore, when commuters complained about the lack of connectivity between some stations, the planners introduced a fleet of feeder buses, which led directly to increased ridership. Another innovation was to design women-only coaches in response to a Delhi government survey which revealed that almost half of female passengers felt unsafe using public transport in Delhi. In fact, the blueprint for the Metro itself reflected demand patterns learned during stakeholder consultations. Clearly, the operating model not only enabled the receipt of feedback but was also flexible enough to adapt and incorporate these innovations.

PERIPHERAL DRIVER 3 **MOTIVATED AND SKILLED WORKFORCE**

The Delhi Metro project could identify, attract and retain high-quality human capital, which contributed to the success of the project. The management focused on hiring people with strong qualifications and plenty of real-world experience, often from India's renowned railway service. The expertise of these railway engineers ensured that the Metro followed existing infrastructure paths, such as the city's radial corridors, through a smart mix of overhead and underground routes that maximized space usage and efficiency.

The project also invested in training and up-skilling its employees, establishing its own training institute and collaborating with the Indian Institute of Technology in Delhi to provide advanced degrees to its employees. An on-the-job employee rotation plan exposed workers to other facets of the Metro and provided them with better growth opportunities. This focus on training and growth, coupled with the knowledge that workers' ideas were treated seriously, instilled a sense of pride and loyalty among the workforce.

HOW ESEVA TAMED A PUBLIC BUREAUCRACY IN INDIA

HYDERABAD, INDIA, WANTED TO CREATE A ONE-STOP-SHOPPING MODEL FOR PUBLIC RECORDS AND SERVICES. ITS SOLUTION, “ESEVA,” HAS SET RECORDS ACROSS INDIA FOR FAST, FLEXIBLE, HIGH-QUALITY CUSTOMER SERVICE.

Whether your life was just beginning, nearing its final season or at different points in between, if you lived in Hyderabad, India prior to 1999, the public bureaucracy was probably impeding your progress in one way or another. Public services such as birth and death certificate registrations, passport applications, and utility bill payments required the input of distinct government departments often separated by jurisdictions, billing mechanisms and procedures. For example, citizens in one district of the city could not make payments in another, making the use of Hyderabad's public services tedious and time consuming. Other hurdles involved an overreliance on paper-based processes, fixed (and inconvenient) working hours, and other rigid office operating policies that lowered operational efficiency and created long waiting times at government offices. The system also lacked adequate transparency and accountability.²⁸

CREATING A ONE-STOP SHOP FOR CITIZENS

To address these bureaucratic challenges, the government created eSeva, a shared-services platform, in 1998 that enabled one-stop shopping for citizens. Envisioned as a template for both urban and rural areas, eSeva follows a BOOT model (Build, Operate, Own and Transfer) that aims to transfer the day-to-day administration of the service centers to private operators.²⁹ Following the initial success of two pilots, eSeva was rolled out with special attention paid to local relevance and uptake.

The program has expanded the services it offers from government-to-consumer (G2C) transactions such as utility payments and taxes, to business-to-consumer (B2C) services like mobile phone bill payments. The number of eSeva kiosks in Hyderabad reached 405 by 2016 and in total exceeded 4,500 across different participating districts beyond the city of Hyderabad.³⁰

In one assessment of the project, nearly 97% of survey participants indicated a preference for eSeva's approach over traditional government systems.

Researchers found that eSeva reduced average waiting times from nearly 33 minutes to under 15 minutes, helping to cut transaction costs.³¹ The project built additional eSeva centers in response to citizen demands for increasingly convenient services and data availability, all reinforced by a series of effective marketing campaigns. And all this boosted government revenues. For example, in April 2002, the service had 60,000 paying customers; a year later under eSeva the total had grown to more than 100,000. The project's success led to the adoption of similar approaches in other states, and eSeva has won a wide variety of awards.³²

Measuring the project's comparative impact, eSeva substantially outperformed another e-governance project in a similar-sized city in a variety of ways. From the number of customers served and transactions processed to the average number of trips a citizen had to make to complete a transaction, eSeva handily surpassed the other initiative.

FACING COMPUTER ILLITERACY, BANDWIDTH ISSUES AND PUBLIC INERTIA

The secret to eSeva's success - gathering all the services from different government agencies and businesses under one roof - turned out to be a complicated and time-consuming challenge. Likewise, expanding the model to more rural areas was difficult because of low levels of computer literacy, poor internet penetration and low-bandwidth connections to servers. Yet another hurdle involved finding effective ways to change the mindsets of thousands of citizens regarding the adoption of technology and the use of eSeva kiosks to make payments.

HOW THEY DID IT

Three core drivers and two peripheral ones interacted to make the eSeva project a success (Figure 9). The core elements included having a proactive political sponsor; the citizen-focused nature of the project's design; and the use of robust monitoring mechanisms. Two peripheral drivers - project leader expertise and the inclusion of business-led operations - reinforced the project's successful dynamic.

Hyderabad, India

PUBLIC BUREAUCRACY eSEVA

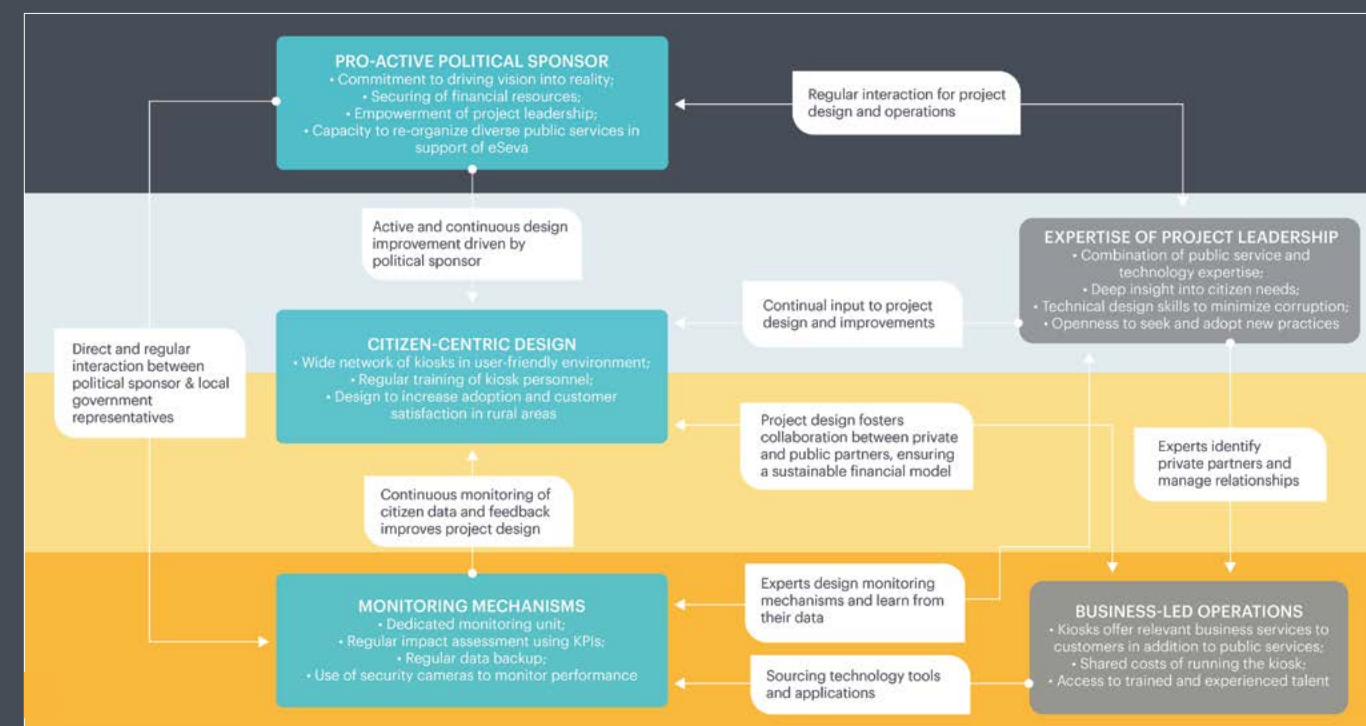


FIGURE 9

The core (blue) and peripheral (grey) drivers of eSeva's success.

- ENVIRONMENT
- LEADERSHIP
- ORGANIZATION
- IMPLEMENTATION
- CORE CONDITIONS
- PERIPHERAL CONDITIONS
- FACILITATING MECHANISMS



CORE DRIVER 1 A PROACTIVE POLITICAL SPONSOR

The eSeva project benefited from the attention and commitment it received from the chief minister of state, who envisioned the initiative as part of a larger program to establish the city of Hyderabad as a major IT and technology hub in India. To that end, he used eSeva to digitize the city's public services, leading by example. The minister had a deep belief that IT could improve governance. He was a pragmatic sponsor, willing to learn from what worked elsewhere, and translating those lessons to his environment. In one case, for example, he borrowed ideas from Singapore regarding electronic governance to make eSeva's service delivery more efficient. His political sponsorship made it possible to combine the required government services from different departments under one roof. What's more, being the chief minister of the entire state reduced issues of accessing finance.

The chief minister was a very proactive sponsor. After carefully selecting and empowering the project leadership, he remained involved through regular interaction on matters of project design and operations. His focus on monitoring and continually improving the project design meant that eSeva evolved to meet genuine citizen needs in a way that was appropriate to the local context. And he made sure to maintain direct and regular interaction with local government representatives and other sponsors, ensuring continual alignment as the project evolved. Robust monitoring mechanisms were crucial to enable this rich, regular feedback.

CORE DRIVER 2 CITIZEN- CENTERED PROJECT DESIGN

The eSeva project placed the customer at the center of its focus, making available water, fans, air conditioners and chairs at the kiosks, for example. Equipped with technological infrastructure, the kiosks increased transparency, clearly displaying prices for services, for instance, and making services more customer friendly. One leadership goal was to treat every customer as a Very Important Person (VIP) and this was taken seriously, with kiosk personnel receiving regular training. Prior to eSeva, interactions with government agencies were often difficult; after it, people said they "felt dignified" when using it.

The project's design was under continual improvement, drawing input from a variety of sources, not least from the continuous monitoring of citizen data and feedback from users themselves. Moreover, eSeva's model encouraged its managers to contribute their own ideas for improving the system. For its part, the project leadership was willing to adapt the project design to incorporate the best of those ideas. Regular video conferencing and physical meetings across management and leadership enabled this back-and-forth of ideas.

Crucially, eSeva's design benefitted from the lessons learned from successes and failures elsewhere. The leadership's experience in both public services and technology brought a treasure trove of such lessons, which they successfully adapted to eSeva's environment. These ideas included employing a token system to reduce queues at the kiosks, keeping the kiosks open after hours to cater to office workers, providing around-the-clock access to some services via the Internet, and remaining open on public holidays.

An example of how eSeva's solutions adapted to the local context is that in more rural zones, kiosk designs provide either digital or manual services, enabling people to print out and "rubber stamp" documents, which led to higher adoption of the services.

CORE DRIVER 3 MONITORING MECHANISMS

The political leadership took a hands-on approach regarding eSeva, which included playing an active role in monitoring the performance of the project. Academics that researched the project say the chief minister clearly described the desired implementation process and then claimed he would monitor progress every week. To that end, the project established a dedicated center to monitor performance via mechanisms such as key performance indicators, budgetary controls and quality assurance audits. Regular backup of data ensured system continuity, and the center also utilized remote security cameras to observe the day-to-day functioning of eSeva kiosks. The project quickly gained a reputation for being more sensitive to feedback than other public service programs in the country. Indeed, it was this robust monitoring system that allowed the service to evolve and become so relevant to its target citizens. It also ensured that regular feedback and reporting were available to maintain the trust and confidence of stakeholders.

PERIPHERAL DRIVER 1 PROJECT LEADERSHIP EXPERTISE

The eSeva project's leadership had a strong understanding of the expectations of the political sponsors and target citizens. The monitoring mechanisms they designed allowed them to keep learning more about the citizens and adapt the design of the initiative accordingly. They offered continual input to project design improvements, which required both an openness to seek and adopt new practices as well as the technical ability to design systems that are appropriate for the local context. For example, the project designed processes to minimize the scope for corruption. The leadership also possessed the technology expertise - both strategic and operational - to achieve their design goals.

Relationship management was an important part of the role for the project leadership. They interacted regularly with the project sponsors and identified key partners from the private sector to facilitate effective operations. Having a deep understanding of conditions on the ground and continual access to performance data made these relationships extraordinarily constructive.

PERIPHERAL DRIVER 2 BUSINESS-LED OPERATIONS

The eSeva project engaged in public-private collaboration at two levels. At the first level, government departments and private companies collaborated with eSeva to offer G2C and B2C services at the kiosks. At the second level, the project collaborated with several leading IT companies such as CMS Computers Ltd. and Ram Informatics Ltd. to manage the kiosks. Private companies benefited from the huge volume and scale of demand brought by the system, while the government benefitted from the sharing of costs and responsibilities of running the kiosks. The private sector also ensured service quality levels. For example, by collaborating with private agencies, eSeva could bring in needed talent and training capabilities to assemble a team capable of executing the project successfully.



LEARNING TO SURVIVE (AND THRIVE) IN SOUTH AFRICA

WITH HIV AND AIDS PREVALENT, AN INNOVATIVE PROGRAM MOBILIZES THE COMMUNITY TO GIVE YOUNG PEOPLE IN KWAZULU-NATAL THE EDUCATION AND GUIDANCE THEY NEED.

Growing up in KwaZulu-Natal (KZN), South Africa, can be very challenging. It has the highest HIV rates in the country,³³ and the young people who live there must cope with the common loss of parents and caregivers to the disease. The resultant disruption of key relationships in their lives makes them fall even further behind socially and economically.

Many local young people lack access to adequate health and social information; material that could change their lives or save them from preventable diseases like HIV and AIDS. Local schools at best offer only very limited information on these topics—not the detailed insights that could make a sustainable difference in students’ lives. Access to digital information sources is rare; government television campaigns do not reach many disadvantaged youths, and radio campaigns rarely offer sufficient depth. Young South Africans need information and in some cases mentors to make decisions that will help them cope with their circumstances and improve their lives.

SAVING LIVES WITH INFORMATION

To rescue the children of KZN, the Isihlangu Health and Development Agency along with the Population Council, a global NGO, developed and piloted a life orientation program. Called Siyakha Nentsha (isiZulu for “building with young people”), the program seeks to improve the lifelong skills and well-being of the community’s youth.

The program initially conducted focus group discussions with guardians and traditional leaders, which led to the widespread inclusion of both girls and boys into the program. Girls were a primary target population because of their vulnerability to HIV and early pregnancy. The program included boys with the goal of building their knowledge and skills. Another goal involved enabling girls and boys to work together, interact socially, learn to respect one another, and consequently move beyond the objectification of the opposite sex. The program thus offered a unique opportunity to measure outcomes for both females and males in a setting where gender relations often become strained.³⁴

CHANGING ATTITUDES

The project focused on three sets of students: a control group; a group that received HIV education, social support and stress reduction instruction; and a third group that received HIV education, social support and financial education.³⁵ Both the streamlined and full versions of Siyakha Nentsha led to important changes in young people’s attitudes and behaviors, though the impact differed between the three groups and between males and females.

For example, compared to the control group, all Siyakha Nentsha participants (regardless of sex or version of the program) were more likely to know where to obtain contraception, reported a large increase in their knowledge of social grant requirements and criteria, gained improved budgeting and planning skills, and were more likely to have attempted to open a bank account.³⁶

Compared to the control group, Siyakha Nentsha girls reported feelings of higher self-esteem and expressed greater confidence in their ability to obtain contraception if necessary. Siyakha Nentsha boys were more likely to have remained sexually abstinent between survey rounds, and Siyakha Nentsha boys who did have sex reported having fewer sexual partners than did boys in the control group. And compared with participants who received the partial Siyakha Nentsha package (i.e., health and social skills training), girls who also received financial skills training felt greater levels of social inclusion in their communities and were more likely to have obtained a national birth certificate. Likewise, Siyakha Nentsha boys who received a financial education were more likely to have landed jobs between survey rounds.³⁷

STARTING FROM SCRATCH

The challenges the program faced included having to start from scratch, persuading educators, parents and communities to participate, and dealing with the lack of adequately trained facilitators, which forced the program to coach and train people. It’s important to note that this project was undertaken in an area characterized by tribal communities, with community structures and cultural norms that project planners needed to understand and respect.

HOW THEY DID IT

Siyakha Nentsha succeeded thanks to the interaction of three core conditions: it had an experienced and credible leader, a citizen-centered design, and a motivated and empowered workforce. The dynamic between these conditions was further reinforced by two peripheral drivers: supportive government sponsors and a highly-engaged citizenry (Figure 10).

Umlazi, South Africa

EDUCATION SIYAKHA NENTSHA

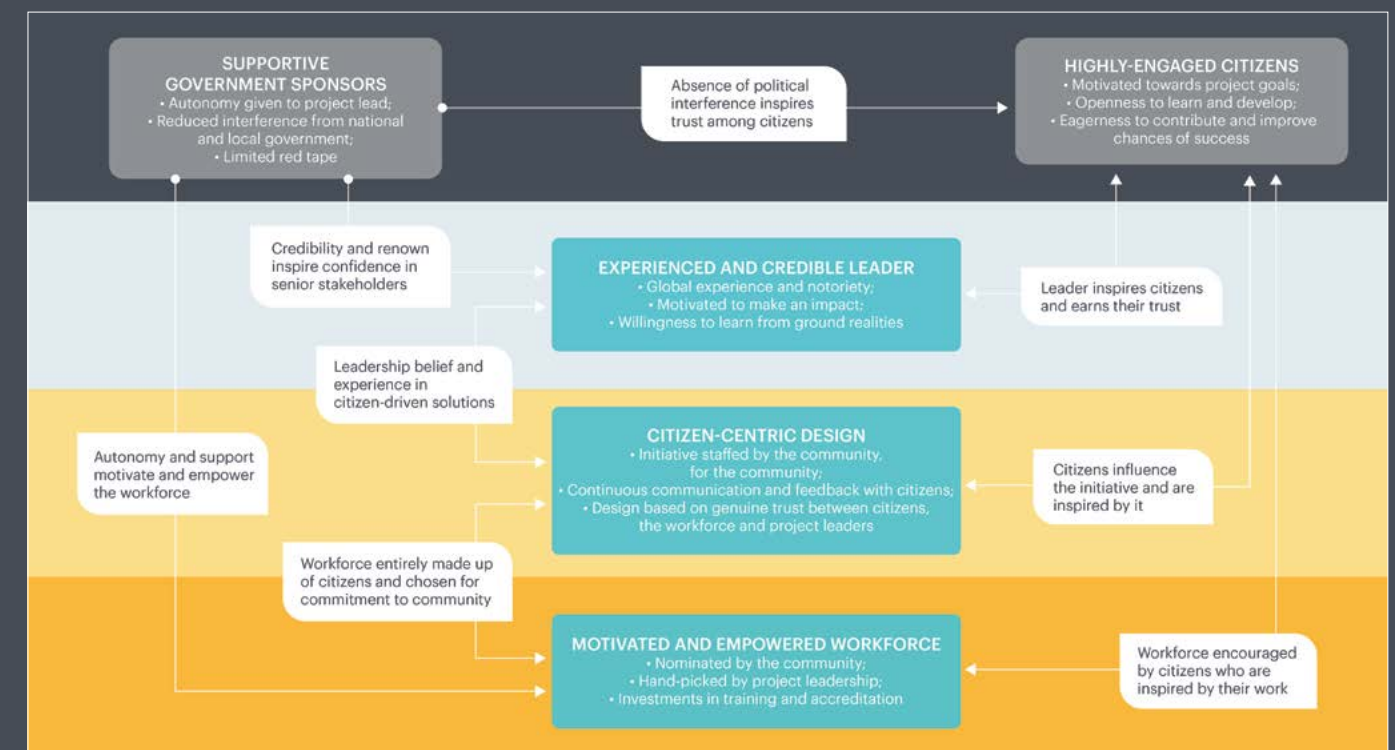


FIGURE 10

The core (blue) and peripheral (grey) drivers of Siyakha Nentsha’s success.

- ENVIRONMENT
- LEADERSHIP
- ORGANIZATION
- IMPLEMENTATION
- CORE CONDITIONS
- PERIPHERAL CONDITIONS
- FACILITATING MECHANISMS



CORE DRIVER 1 **EXPERIENCED, CREDIBLE LEADERSHIP**

The Siyakha Nentsha program benefited from highly-experienced leadership in the implementation team, who had expert skills in a variety of relevant fields, from HIV prevention to project evaluation methodologies. The team also drew from an extensive and extremely helpful network of relationships within the community, including among community leaders.

At the top, Kelly Hallman, the project's managing director, enjoyed high levels of visibility with the senior leadership of the government education department, which helped to set up the program. Indeed, they set up a new, independent NGO, dedicated to this program. Under Hallman's leadership, the team was highly motivated to leverage lessons and experience from around the world, but to design solutions that were ultimately driven by the local realities on the ground. This vision and strong belief in citizen-driven solutions sat at the core of the program's approach.

CORE DRIVER 2 **A CITIZEN- CENTERED DESIGN**

Given a free hand to implement the project as they chose, the project leaders pursued an extremely inclusive approach, listening to all valid opinions, especially those from the community, the students and the facilitators, and incorporating important viewpoints into the program. Before designing the project, leaders held multiple meetings with different parts of the community, such as the youth, care givers, the elderly and community leaders. Consequently, all involved felt that their opinions mattered, and that they had contributed to the success and delivery of the program, as well as its selected content.

This meant the project was based on genuine trust between the citizens, the workforce and the project leaders. This in turn motivated and empowered all these stakeholders. Moreover, the continuous interaction with the citizens meant the project evolved to be highly relevant to local realities. In fact, motivated members of the community ultimately entirely staffed the project, a crucial characteristic that meant locals really viewed it as an initiative by the community, for the community, inspiring tremendous engagement and support from the citizenry.

CORE DRIVER 3 **A MOTIVATED AND EMPOWERED WORKFORCE**

The leaders of the implementation team worked hard to train the facilitators so that they were highly capable and could achieve effective results with the students. This required significant investment in training as the facilitators began with very little, if any, experience. In fact, the project team registered the facilitator course via the South African Qualifications Authority (SAQA), to provide them with a meaningful credential at the end of their training. Leaders viewed this as a reward for the facilitators, who earned only basic salaries during the course administration.

A crucial mechanism driving the positive dynamic between the project and the community was that the workforce entirely consisted of citizens, chosen for their commitment to the community. The tribal community initiated this mechanism and the project leadership understood the valuable motivation and engagement it could generate. This dynamic also enabled the project to evolve more closely towards understanding and fulfilling local needs.

PERIPHERAL DRIVER 1 **HIGHLY ENGAGED CITIZENS**

Community involvement was key. It helped that many of the would-be facilitators were jobless and viewed the offered positions not only as a very important opportunity for their communities but also as an opportunity for themselves. What's more, the fact that the facilitators lived in the community gave the project a permanent presence, voice and ears within its target group.

To capitalize on this relationship, the project planners held regular community meetings throughout the project implementation with elders and leaders. They also constantly networked in the communities, ensuring strong endorsement from the local citizens, commitment from local leadership and ultimately, highly relevant project outcomes.

PERIPHERAL DRIVER 2 **SUPPORTIVE GOVERNMENT SPONSORS**

The government's support and sponsorship for this initiative was felt through its efforts to reduce political interference and red tape, and to give autonomy to the project leadership in designing and implementing their plans.

The absence of political interference was important in inspiring trust among the citizens, and the autonomy was crucial in empowering and motivating the workforce. The government sponsors themselves felt assured in allowing this autonomy because of their trust and confidence in the project leadership.



WATER WORKS: HOW ETHEKWINI QUENCHED ITS THIRST

A SOUTH AFRICAN MUNICIPALITY COLLABORATED WITH ITS OWN CITIZENS TO DEVELOP AN AWARD-WINNING PLAN THAT PROVIDES WATER AND SANITATION SERVICES.

South Africa's eThekweni is the largest municipality in the coastal province of KwaZulu-Natal, and the third-largest in the nation, with 3.4 million people in 2011. eThekweni is a sophisticated city, but contains many informal settlements that house the poor.³⁸ Created in 2000 when the city of Durban incorporated the rural areas surrounding it into the larger municipality, eThekweni faced numerous challenges, with a lack of adequate water supplies and sanitation service ranking high on the list. Roughly two-thirds of the approximately three million people in the municipality had to make do either with poorly maintained, dilapidated water and sewerage services, or in many cases no services at all.³⁹

FULFILLING A CONSTITUTIONAL MANDATE

South Africa's 1996 constitution guarantees citizens the right to potable water, and puts the responsibility on local government to deliver this service.⁴⁰ With continued population growth and a cholera outbreak⁴¹, authorities felt compelled to address the severely strained water and sanitation supplies. Led by eThekweni Water and Sanitation, the solution they developed was both innovative and highly successful.

The eThekweni Water and Sanitation project succeeded on multiple levels, winning several awards both locally and at the international level. For example, the Stockholm International Water Institute has recognized Durban for having one of the world's most "progressive water utilities," with its open approach to experimentation and piloting of new solutions.⁴²

By 2013, more than 92% of households had basic water and almost 77% had basic sanitation.⁴³ In the 12 years up to 2015, more than 1.3 million people gained access to piped, potable water and over 700,000 received new sanitation services. The project has boosted free basic water supplies from six kiloliters a month to nine. Well-lit communal toilets were built in dense, informal settlement areas and cleaned regularly by toilet attendants, generating cleanliness satisfaction scores of over 80% (compared to 50% in a research control group).

LEAKS, A BOOMING POPULATION, FUNDING LIMITS, RESPECTING ZULU CULTURE

The project faced several critical challenges along the way. The water supply was intermittent due to leakages combined with numerous illegal connections, system tampering and outright theft. The city also suffered from uncontrolled and unplanned population increases in both rural and urban areas, as well as frequent electricity outages due to excessive load. At the same time, the project had to manage the expectations of citizens in low-income areas, and deal with a variety of dysfunctional institutions.

At the project level, limited funds forced the municipality to make tough tradeoffs between expanding existing infrastructure and maintaining existing assets. The project also needed to build trust-based relationships with local communities and to accommodate the local Zulu culture, which required both time and effective communication strategies. Finally, the initiative faced several engineering hurdles, since the rural areas surrounding eThekweni were hilly and uneven, requiring pipeline design innovations to ensure an efficient water supply.

HOW THEY DID IT

The success of the eThekweni project depended on the interaction of three core drivers and two peripheral ones (Figure 11). Core elements were having an innovation-focused leader, adopting a rigorous project management approach, and fostering a collaborative implementation style. Peripheral drivers included supportive political sponsors and strong engagement with the community. Genuine collaboration sat at the heart of the initiative's design, with the intimate involvement of the community, as well as a series of expert organizations. Robust monitoring, reporting and feedback loops across stakeholders ensured that the project remained relevant to its local context and stayed focused on achieving outcomes.

eThekweni, South Africa

URBAN PLANNING WATER AND SANITATION SUPPLY

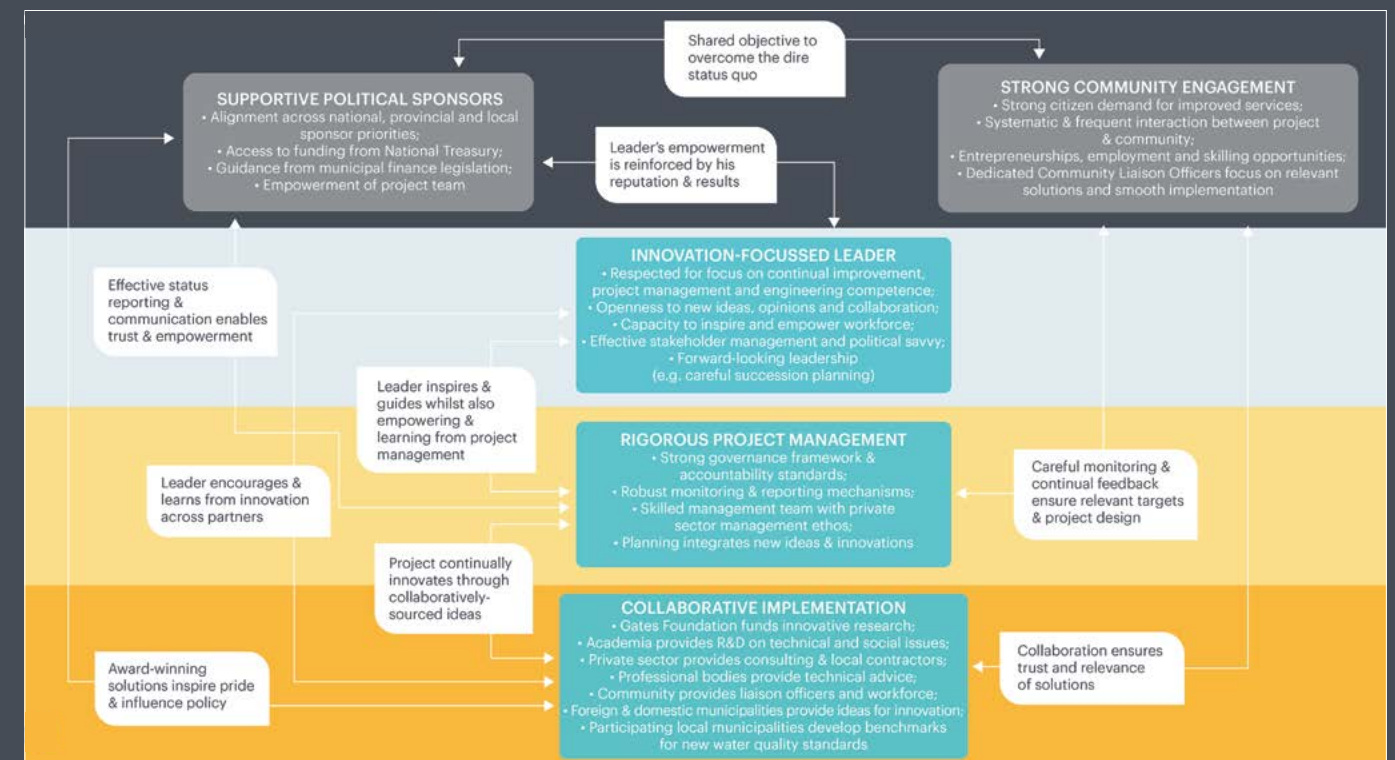


FIGURE 11

The core (blue) and peripheral (grey) drivers of eThekweni's success.

- ENVIRONMENT
- LEADERSHIP
- ORGANIZATION
- IMPLEMENTATION
- CORE CONDITIONS
- PERIPHERAL CONDITIONS
- FACILITATING MECHANISMS



CORE DRIVER 1 AN INNOVATION- FOCUSED LEADER

The leader and leadership team had a very clear vision for this project and held regular meetings to make sure everyone was on the same page. The head of the eThekweni Water and Sanitation Department was well respected for his focus on continual improvement, rigorous project management and for his engineering expertise. He gave the project team the freedom to innovate, encouraging the workforce to collaborate with the local university as well as conducting their own innovations in tandem. He allowed staff members to participate in professional bodies to increase their skills and their relationships with other knowledgeable engineers in the field. He also maintained an open-door policy that enabled insightful communication between the project team and the stakeholders. In doing so, the leader was a strong source of inspiration for the workforce, and the resulting enthusiasm paid dividends through real innovations and strong outcomes. These successes in turn further reinforced support and commitment for the leader and for the model.

Participants said people felt that their opinions mattered and that they were given the chance to voice their ideas. The project leader showed a genuine concern for the sustainability of the project, investing in careful succession planning and shielding the team from any political interference, giving them the opportunity to focus exclusively on their jobs while he took care of political issues.

CORE DRIVER 2 RIGOROUS PROJECT MANAGEMENT

The leadership team held the people running the project accountable for their actions. Project management was professional and focused on efficiency and transparency. For example, every phase of the project was carefully monitored, while a City Ombudsman's office investigated any queries regarding possible irregularities. All participants had to complete a gift register to mitigate any issues concerning corruption or undue influence over contracts. The project required full disclosure as well as strict adherence to the Municipal Finance Management act (MFMA). It did not tolerate undesirable practices and made suggestion boxes available so that people could anonymously report possible fraudulent transactions or criminal behavior. What's more, everyone involved in the project had to report all work-related activities outside of their jobs. Consequently, the entire project management process dealt stringently and rigorously with accountability and corruption issues.

The focus on monitoring, transparency and accountability generated trust among stakeholders, enabling more autonomy for the workforce to innovate in collaboration with partners. Indeed, the project management style encouraged the adoption of new ideas and innovations from across the project's collaborative ecosystem.

CORE DRIVER 3 COLLABORATIVE IMPLEMENTATION

Participants characterized the project as being intensely collaborative, with all parties exhibiting a desire to succeed and a willingness to innovate continuously. Good ideas were welcomed from everywhere. For example, citizen input on the regular cleaning of toilets ultimately resulted in a formal policy within the initiative.

The local municipality collaborated with other municipalities to develop their collective engineering and water and sanitation skills. The project also worked with water research organizations in South Africa to develop a national benchmark initiative, and successfully partnered with the private sector for much of the construction effort.

PERIPHERAL DRIVER 1 SUPPORTIVE POLITICAL SPONSORS

Partnerships with professional bodies meant that the project team gained further training and advice from skilled national experts.

The Water and Sanitation Department and the project team collaborated successfully with a local university to develop the ventilated pit latrines and dry feces processes, leading to additional innovation and understanding of the social aspects of using the latrines in the local communities. The project team partnered with organizations such as the Bill Gates Foundation, which had invested in dry-toilet research. It also used other international projects as benchmarks – liaising with the project teams and learning from them.

Team leaders recruited and trained community liaison officers from the local unemployed population who went out to the local communities to train people regarding the use and maintenance of the new pit latrines. Before building the latrines, they sought advice from the communities regarding where to site them and other considerations to ensure their optimum use. It took at least five visits from the community liaison officers before the final installation of each toilet, which opened frequent opportunities for communication and learning—a process made easier by the fact that the project employed local people in the construction teams to build the basic toilet blocks.

The constitutional requirement for the supply of basic water and sanitation falls under the Water and Sanitation Pillar of South Africa's National Development Plan (NDP). As a result, the national government supported this project by making funding available through the National Treasury. Without this support, the local government would have lacked sufficient funds to carry out the project. The faith placed by the sponsors in the leader resulted in innovative solutions, tangible outcomes and global recognition, which inspired pride and yet-stronger commitment among the sponsors.

Even the sponsorship role was collaborative, as government departments such as the Environmental Department, national, provincial and local housing departments as well as town planning departments worked together successfully to bring the project to fruition. At the same time, the City Manager of eThekweni gave the project team room to make decisions, which fostered a relationship based on mutual trust and understanding.

PERIPHERAL DRIVER 2 STRONG COMMUNITY ENGAGEMENT

Active communication between the project team and participating communities was a crucial factor of this project's successful dynamic. For example, discussion topics included the process for the dry-toilet installations, as well as timing and other considerations that would affect the local community. The project team made sure it secured buy-in from the communities, and that all parties had the chance to voice their opinions and ideas on the placement of the toilets. Likewise, the recruitment and training of community liaison officers, drawn largely from among the unemployed people from the communities, helped with this process of socialization and skills transfer. The project also used local contractors for the creation of new infrastructure, thus creating local jobs, and small businesses benefited from the increased activity in the areas. This further sparked the committed involvement of the local communities, ensuring their buy-in to the overall project.



MYCITI: CAPE TOWN BUILDS A SAFE AND SUCCESSFUL BRT

CITIZENS IN CAPE TOWN, SOUTH AFRICA, NEEDED A SAFER, MORE RELIABLE MASS TRANSIT SYSTEM. THE CITY'S NEW BUS RAPID TRANSIT SYSTEM OFFERED THAT, AND A LOT MORE.

Cape Town, South Africa's second largest city, saw its population grow 45% from the mid-1990s to reach 3.7 million people in 2011. The resulting boom had city planners looking for safe, affordable public transportation options that were also efficient and dependable.⁴⁴ While the city already had train, bus and taxi services that logged over a million trips a day, the safety and reliability of these services had deteriorated over time. Meanwhile, the flood of private cars on city streets was causing increasing congestion and delays, and city planners knew they couldn't continue to build new roads to handle the increasing number of vehicles.⁴⁵

STRATEGICALLY INVESTING IN A BRT SYSTEM

In response to these challenges, the city strategically invested in establishing a bus rapid transit (BRT) system, which would also fulfill part of Cape Town's commitment towards becoming the host city for the 2010 FIFA World Cup. Pioneered in Brazil, BRTs typically operate on dedicated roadways and provide quick transit through congested urban areas that is reliable, safe and cost efficient. The project had an explicit goal: to offer over 80% of citizens access to public transportation no more than 500 meters from their homes. Another mandate was the use of a "one ticket, one timetable" system in which passengers purchased one ticket and could then switch between different modes of public transport (e.g., buses, trains and taxicabs).⁴⁶

INTEGRATING PUBLIC TRANSPORTATION

Called MyCiTi, Cape Town's BRT became the cohesive glue behind the city's vision of an integrated public transportation system. With millions of passenger journeys completed on an ever-growing network of routes, the social impact has been tremendous. Passengers can now make use of a safe, scheduled, reliable, and comfortable bus service to move around the city.

A website accessible via mobile phone provides passengers with real-time information on bus departure times for every stop on every route. All stations feature passenger information displays and customers can buy transit passes at MyCiTi stations or at selected retailers.

MyCiTi fares are comparable with those of minibus taxis or other buses, but the service it provides is safer, more reliable and more comfortable. The introduction of dedicated BRT infrastructure, coupled with upgraded pedestrian and cycling environments, have not only revolutionized the urban environment but also the public perception of an inclusive and sustainable roadway.

In doing so, the initiative has helped respond to apartheid-era planning that kept some communities distant from economic, educational and recreational opportunities. At the same time, the project trains former taxi and conventional bus drivers to operate MyCiTi buses, thus re-skilling the workforce.

The timeliness of buses compared very favorably with the Transport for London standard of having 85% of buses on-schedule,⁴⁷ and passenger feedback was overwhelmingly positive.⁴⁸

The bus system and the pre-paid gateway ticket known as MyConnect have gained international recognition as leading examples of service excellence.⁴⁹ Today, other South African cities study the MyCiTi project as inspiration for their own bus rapid transit systems.

MANAGING PUBLIC REACTION AMID DISRUPTION

Local authorities in South Africa had no experience in planning and implementing BRT infrastructure projects. Beyond technical challenges, public opinion had to be managed while establishing dedicated bus lanes in existing road systems, expropriating land, complying with environmental regulations and building stations in the middle of existing traffic.

HOW THEY DID IT

The MyCiTi project succeeded thanks to the interaction of three core and two peripheral drivers (Figure 12). The core drivers reinforcing one another were the engaged ecosystem of partners, the collaborative organizational model and the motivated and capable workforce. The two peripheral drivers strengthening the successful dynamic were strong community engagement and committed political sponsors.

City of Cape Town, South Africa

TRANSPORT BUS RAPID TRANSIT (BRT) - MyCiTi

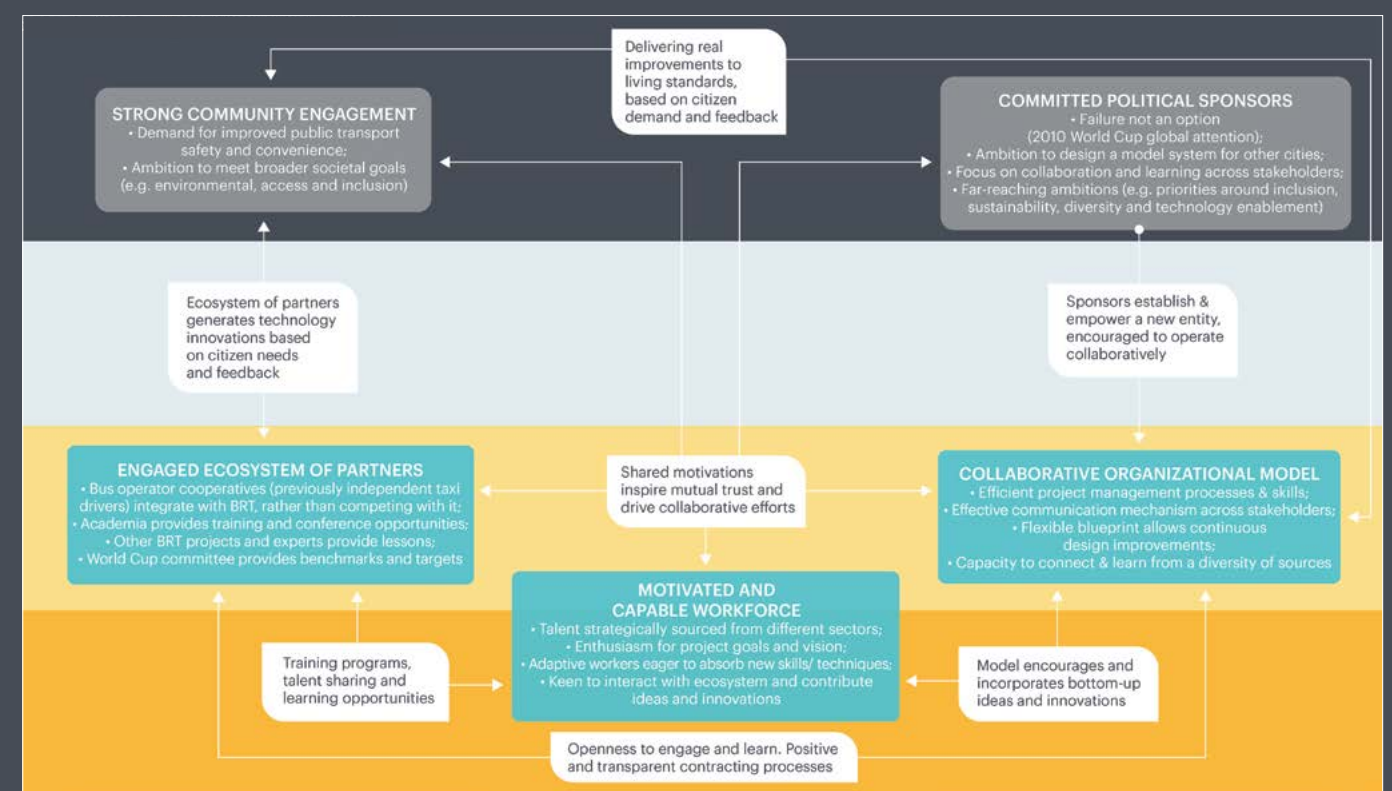


FIGURE 12

The core (blue) and peripheral (grey) drivers of MyCiTi's success.

- ENVIRONMENT
- LEADERSHIP
- ORGANIZATION
- IMPLEMENTATION
- CORE CONDITIONS
- PERIPHERAL CONDITIONS
- FACILITATING MECHANISMS



CORE DRIVER 1 AN ENGAGED ECOSYSTEM OF PARTNERS

The project benefited from strong collaboration and buy-in among stakeholders, all highly engaged in discussions and willing to learn from each other. The resulting collaborative ecosystem had a strong sense of innovation – a desire to push the boundaries and to create something extraordinary. For example, the project team negotiated with mini-bus taxi operators to create Bus Operator Cooperatives that integrated with the BRT system, rather than competing against it.

The University of Cape Town played a crucial role in developing courses for public transport engineers and became a fertile ground for discussion and innovation among peers. These opportunities to share and learn not only improved the capabilities of the workforce but also motivated them to innovate and learn yet further. The inspirations for many new innovations were sourced from the workforce, citizens and other members of the ecosystem.

The project team visited quite a few other BRT projects, which put it in contact with global experts, knowledgeable peers and external consultants. World Cup Committee experts also took part in planning the BRT. A defining element of this project is that it molded a lot of experience and expertise into the project team. This was thanks to a shared commitment toward the same goals and an organizational model that allowed these disparate groups to effectively interact and shape the outcomes together.

CORE DRIVER 2 A COLLABORATIVE ORGANIZATIONAL MODEL

MyCiTi benefitted from political sponsors that encouraged a collaborative model to benefit from the widest array of experts possible. They established and empowered a dedicated team of highly-skilled managers with efficient project management processes to implement this model. Characterized by clear communication and transparent contracting processes, the project kept a diverse collection of stakeholders connected via effective communication channels.

These connections allowed for shared learning, skill transfer and most important, the generation of new ideas and innovations. Critically, the project blueprint was flexible enough to allow suitable innovations to be incorporated, enabling a flow of continuous improvement as the project evolved. This in turn stimulated the workforce and ecosystem partners as they could see their input making a tangible difference to the project.

CORE DRIVER 3 A MOTIVATED AND CAPABLE WORKFORCE

Project leaders engaged with the workforce in a collaborative environment that motivated them to “know more and do better.” Inexperienced workers learned the ropes from extremely capable and skilled engineers. What’s more, the participation of a local university motivated workers to increase their skills and understanding of public transport issues.

The workers themselves came from a variety of sectoral backgrounds, enhancing learning opportunities across the group and increasing the chances of innovation through the cross-pollination of ideas. They were eager to interact with the ecosystem and contribute ideas, and at the same time, they were adaptive and able to absorb new skills and techniques. Moreover, they were united by their enthusiasm to contribute toward this high-profile project that promised to make a dramatic improvement in people’s lives.

PERIPHERAL DRIVER 1 STRONG COMMUNITY ENGAGEMENT

Continuous engagement with Cape Town citizens during the planning and implementation phases kept the community involved and aware that the project was hearing their requests and pursuing their demands. The community had a voice in specific elements of the BRT system such as the environmentally-friendly bus stops, the buses capable of handling wheelchairs, and the blind-sensitive curbing and pavements. Moreover, the design and innovation processes were always informed and guided by citizen needs and convenience, hence the mobile timetable applications and streamlined, technology-enabled ticketing solutions. The active engagement of the citizenry and the passenger-focused innovation directly led to the high usage of the BRT system.

PERIPHERAL DRIVER 2 COMMITTED POLITICAL SPONSORS

With the eyes of the world on the city for the World Cup tournament, failure was not an option for the political sponsors. A change in the city’s political leadership worked to the BRT project’s advantage, giving the new political administration a strong motivation to make it a successful example of the kind of positive change they wished to symbolize. Beyond improving transportation, the administration saw the project as a route to achieving more ambitious and far-reaching goals, such as social inclusion, sustainability, diversity and technology-enablement. They understood the scale of the challenge and the diversity of skills and knowledge required to achieve it, hence their encouragement for an open and collaborative approach to design and implementation.

The political commitment was visible: Transport for Cape Town, a newly established public transport body, provided the official leadership for this project, headed by the Transport Commissioner. At the same time, a Mayoral Committee Member for Transport served as the political face of the project, handling communications regarding it and other public transport issues in Cape Town.

The National Department of Transport promoted the BRT concept as part of a drive to transform the face of public transport across South Africa. This provided the local Cape Town initiative with sufficient funding to start the process of planning and initial implementation. It also reduced the fundraising pressures that hamper the mobilization of major infrastructure projects.

BREAKING NEW GROUND FOR SOUTH AFRICA'S SLUM DWELLERS

A LAND TITLING PROGRAM IN SOUTH AFRICA BEGINS TO RIGHT SOME OLD WRONGS THROUGH A SHARED COMMITMENT TO TURN SLUM DWELLERS INTO HOUSEHOLDERS.

Polokwane, the capital of the Limpopo province in South Africa, has several major slums at its edges - remnants of the country's apartheid past - filled with illegal or informal squatter settlements. The end of apartheid brought a new urban policy with a variety of legislation aimed at improving the lives of slum dwellers. One result is the rapid urbanization now taking place as cities across the country experience high population growth, accompanied by congestion, deteriorating environmental quality and the increasing cost of urban services.⁵⁰ For example, Polokwane saw its population grow from 508,277 in 2001 to 628,999 in 2011. Disteneng, an informal settlement situated on the outskirts of Polokwane, has a majority black population and was the focus of the project described here.

REPLACING SHACKS WITH STURDY HOUSING

The National Department of Human Settlements (NDOHS) initiated the Upgrading of Informal Settlements Program (UISP) under a broader policy known as, "Breaking New Ground." The program aims to provide tenure security, health and safety, and empowerment to poor residents. An important aspect of this includes the recognition and formalizing of tenure rights. Where it is not possible to upgrade existing informal settlements, as a last resort the program relocates communities. Local political leaders decided that relocation offered the best option for the Disteneng area.

The planners decided to take a comprehensive approach that went beyond simply constructing houses, preferring to build actual communities. This involved the creation of an entirely new settlement with roads, services and facilities, relocating the informal settlement dwellers from Disteneng to a greenfield site approximately 1 to 2 kilometers away. In the first phase, the project successfully moved a thousand households.

NDOHS research conducted with technical assistance from the World Bank revealed dramatic improvements in the lives of participants, from the quality of their dwelling structures (e.g., literally building cement and brick walls, concrete floors and windows where there were none) to the availability of modern lighting and cooking technologies (e.g., shifting from paraffin to electricity). More households now feature traditional landlord/tenant relationships, which boosts stability. Satisfaction levels among those relocated has climbed substantially, especially regarding issues such as access to schools and water quality. Home and settlement safety perceptions also increased significantly.

MANAGING WITH INSUFFICIENT FUNDING

The project did not have the funds to relocate the entire Disteneng informal settlement simultaneously. Planners therefore had to focus efforts on close communication with the community in order to explain the phased approach that they chose and to minimize any perceptions of unfairness in selecting resettled households.

HOW THEY DID IT

The land titling program succeeded thanks to the interaction between three core conditions and two peripheral ones. The core drivers were supportive and committed political sponsors, a citizen-centered project design, and rigorous and transparent project management (Figure 13). The peripheral drivers were in strong alignment with the National Development Plan and a citizen-centered implementation approach.

The inherent upheaval of people's lives demanded citizen engagement throughout the process, and this was enabled through a focus on monitoring, communication and transparency.

Polokwane, South Africa

PUBLIC BUREAUCRACY LAND TITLING PROGRAM

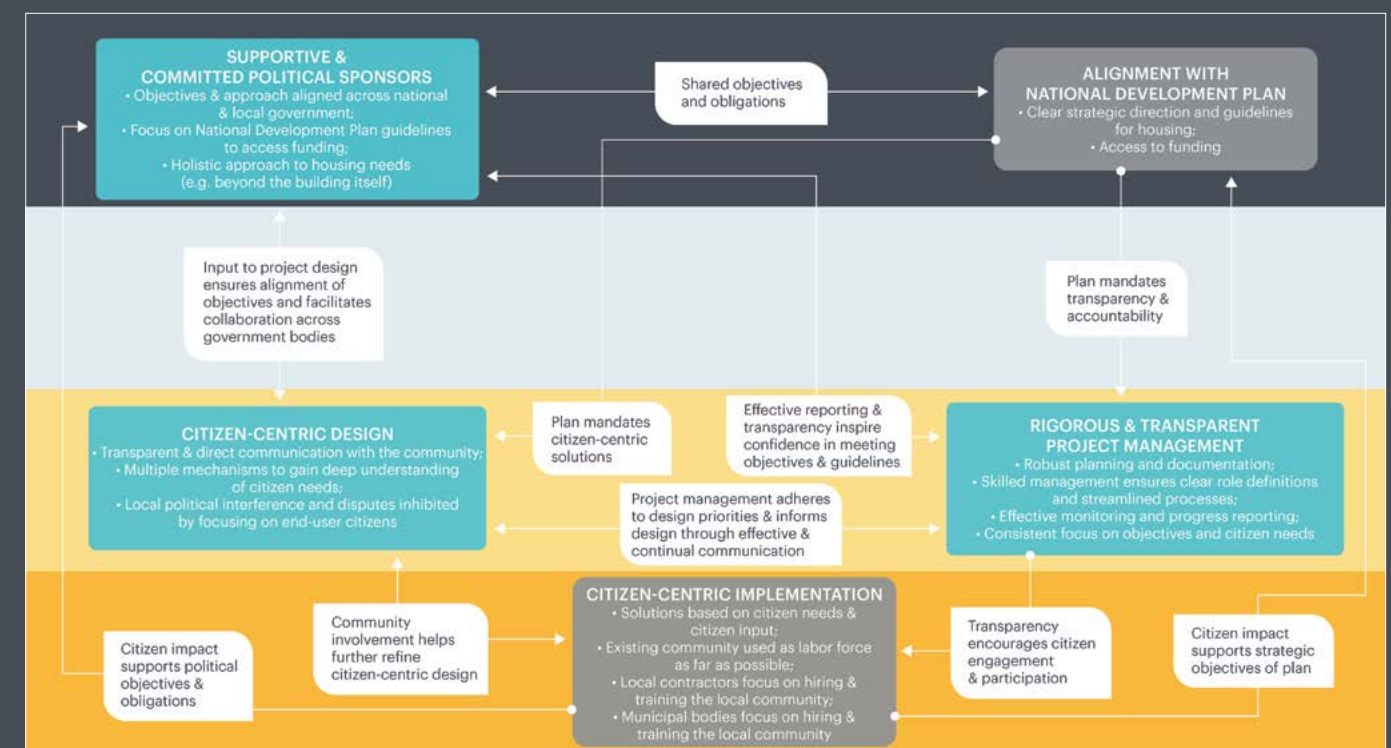


FIGURE 13

The core (blue) and peripheral (grey) drivers of Polokwane's success.

- ENVIRONMENT
- LEADERSHIP
- ORGANIZATION
- IMPLEMENTATION
- CORE CONDITIONS
- PERIPHERAL CONDITIONS
- FACILITATING MECHANISMS



CORE DRIVER 1 SUPPORTIVE POLITICAL SPONSORS

Project leadership had a clear vision and consulted regularly with sponsors, councilors and the involved communities. As a result, there was strong alignment across the various government bodies involved, including the National Development Plan, which was critical to unlock the necessary funding. Moreover, the sponsors envisioned a holistic solution that saw “households” as much more than just “houses”, bringing a thousand new homes together with the complete urban development of streets and government facilities, such as schools, clinics and police stations. The complexity of this task led the sponsors to encourage the development of creative solutions that met these real needs of the affected citizens. The interaction between sponsors and other stakeholders was continuous, ensuring a common understanding of objectives and obligations. The sponsors truly were a core part of this successful dynamic.

CORE DRIVER 2 CITIZEN- CENTERED PROJECT DESIGN

The project was designed with citizen needs at its heart. The project leaders and workforce consulted extensively with involved communities, collaboratively refining their objectives to meet local conditions and requirements. The design therefore had to prioritize processes for transparent and direct communication with the community to drive real cooperation. Clarity and transparency were especially crucial as not every household would be resettled.

The design allowed for the local municipality itself to establish and facilitate feedback processes with the community, building trust between these groups. And by maintaining a strict focus on the end-user citizens, the project design avoided interference from potentially troublesome political voices.

CORE DRIVER 3 RIGOROUS AND TRANSPARENT PROJECT MANAGEMENT

The sheer scale and complexity of the project constituted a major infrastructure challenge that required a robust project management framework. Among other things, rigorous processes and monitoring prevented people from jumping the queue or “buying” their way onto the list to receive a new home; careful record-keeping ensured documentation was available to support claims and defend against accusations; and regular surveys and monitoring guaranteed fairness in the allocation of housing while also reassuring those in the system of their status. The project managers were highly skilled, maintaining clear role definitions, streamlined processes and effective progress reporting.

The rigor of the project management was crucial in inspiring a spirit of openness and transparency across stakeholders. In fact, the National Development Plan mandated open and transparent processes; but a mandate is one thing, and the ability to establish and operate such systems is quite another. The focus on transparency demanded significant capabilities and commitments from the workforce, and their success generated strong confidence and trust among their stakeholders. The citizens, who were in a vulnerable situation, appreciated receiving clear communication about the future of their households; and political sponsors appreciated the regular reporting on progress toward their objectives and guidelines.

PERIPHERAL DRIVER 1 CITIZEN- CENTERED IMPLEMENTATION

The affected communities played a crucial role in implementing the project. The initiative focused on recruiting unemployed workers from the community to construct some of the basic infrastructure. The project supplied training and skills to these citizens, leaving them with improved employment prospects.

Since it was such a large infrastructure project, the relevant local government departments could not handle all the work and thus outsourced much of it to reputable construction and other companies, generating further skills transfer opportunities.

Local municipalities worked hard to establish strong communication channels with the local community. They facilitated communication among the community, local councilors and relevant committees on the project, paying attention to who was included and who was excluded, to ensure a positive and constructive dialogue.

This allowed the implementation to remain focused on real citizen needs, even as those needs evolved.

PERIPHERAL DRIVER 2 ALIGNMENT WITH THE NATIONAL DEVELOPMENT PLAN

The project benefited from the keen desire of the government to fulfill the National Development Plan (NDP) for South Africa and the “Breaking New Ground” policy. The NDP sets out strategic guidelines regarding what the government needs to deliver and what citizens can expect. This includes “decent” housing opportunities for all citizens.

The project planners made wise efforts to align their objectives and those of their sponsors and other stakeholders with those of the NDP. But the alignment went beyond objectives to include a shared vision of the approach to implementation. Specifically, the NDP’s guidelines and mandate for transparent and open approaches provided an important stimulus for the project design, and the project’s adherence to these rules unlocked essential funding.



APPENDIX: RESEARCH METHODOLOGY

PROJECT SELECTION

Following an extensive literature review and the development of our basic theoretical model, our methodology was grounded in the in-depth assessment of public service projects that could be qualitatively and quantitatively evaluated. To ensure a balanced and comparable sample, we chose municipal-level projects in three emerging markets (Brazil, India and South Africa) and four different sectors (education, transport, urban planning, and public bureaucracy). For each country and sector, we identified one project with strong evidence of positive social impact (12 successes), as well as one project with strong evidence of no positive social impact (12 failures), with 24 projects in total.



Successful cases were identified based on counterfactual evidence of positive social value^{51,52,53}. This required comparison with a control group, estimating what would have happened if the project had not been implemented⁵⁴.



Unsuccessful cases were shown to have no positive social value, usually characterized by discontinuation, cost overruns, poor implementation and inappropriateness to local conditions.

INTERVIEW PROCESS

Data was collected through in-depth qualitative interviews anchored to customized rubrics. The rubrics were designed with carefully crafted definitions for each point on the assessment scale, across dozens of assessment criteria. This allowed greater specificity, clarity and comparability than a simple Likert scale, or equivalent^{55,56}. The rubric criteria were categorized beneath four broad headings: environment, leadership, organizational structure and implementation.

At least three interviewees were selected for each project, from former project managers, employees or academics involved at the time of the project's implementation. The fieldwork therefore consisted of more than 70 interviews.

DATA RELIABILITY TESTS

Reliability tests were conducted to mitigate against common respondent biases or errors across interviews and interviewees⁵⁷:

- α** Cronbach Alpha tests were used to check the internal consistency of the interview results. All cases show Alpha scores higher than the acceptable threshold of 0.7;
- K** Kappa agreement tests were conducted against the null hypothesis that responses across raters were randomly determined. In all cases, the corresponding p values allow us to reject the null hypothesis that response scores were randomly chosen.

FUZZY-SET QUALITATIVE COMPARISON ANALYSIS

The fuzzy-set qualitative comparative analysis (fsQCA) employs Boolean algebra to find combinations of conditions consistent with each outcome. The model takes the interview data as input, and as output it produces the specific combinations of project conditions that are statistically likely to result in success, as well as those statistically likely to result in failure. The “fuzzy set” approach allows us to analyze different degrees of presence of specific project conditions, rather than only considering the binary options of their presence or absence. Details of our fsQCA analysis and results are being published in a separate academic paper that focuses on the role of different and multiple collaborative models across the public, private and civil society sectors. These insights informed the findings about the role of collaboration highlighted in this report.

VIRTUOUS-CYCLE DYNAMICS

Based on the principles of systems thinking, we developed diagrammatic representations of the virtuous-cycles driving each successful project. The creation of these diagrams involved an iterative process across the core project team and local teams that were involved in the fieldwork, sometimes also including interviewees that were involved in the projects on the ground. The diagrams were continually refined and their robustness strengthened through cross-validation across the interviewees and secondary research materials.



ABOUT **ACCENTURE**

Accenture is a leading global professional services company, providing a broad range of services and solutions in strategy, consulting, digital, technology and operations. Combining unmatched experience and specialized skills across more than 40 industries and all business functions – underpinned by the world’s largest delivery network – Accenture works at the intersection of business and technology to help clients improve their performance and create sustainable value for their stakeholders. With approximately 394,000 people serving clients in more than 120 countries, Accenture drives innovation to improve the way the world works and lives. Visit us at www.accenture.com

ABOUT **ACCENTURE RESEARCH**

Accenture Research shapes trends and creates data-driven insights about the most pressing issues global organization face. Combining the power of innovative research techniques with a deep understanding of our client’s industries, our team of 250 researchers and analysts spans 23 countries and publishes hundreds of reports, articles and points of view every year. Our thought-provoking research – supported by proprietary data and partnerships with leading organizations such as MIT and Singularity – guides our innovations and allows us to transform theories and fresh ideas into real-world solutions for our clients. Visit us at www.accenture.com/innovation-architecture

ABOUT **INSPER**

Insper is an independent, nonprofit institution dedicated to education and research in the fields of Business Administration, Economics, Law and Engineering. Its mission is to be a leading center and to explore the complementarities of these fields of knowledge. Insper’s teaching activities offer programs for various stages of a career: Undergraduate (business Administration, Economics and Engineering), Graduate and Research Degree (Certificates, MBAs, Law, Professional Masters and Doctoral) and Executive Education (Open-Enrollment programs and Custom programs for companies). In knowledge generation, Insper works through endowed chairs and research centers, which bring together researchers to work on studies and projects focusing on public policy (CPP), finance (CeFi) and business (CENeg). Insper also has centers to foster entrepreneurship (CEMP) and leadership and innovation (CLI). Insper’s academic quality is certified by AACSB, AMBA and Anamba.

ABOUT **INSPER METRICIS**

Insper Metricis is a research center focused on investments and business strategies for high socio-environmental impact. Special emphasis is given to develop tools to assess and reward for impact, as well as validate impact assessments performed by organizations and governments. The center’s key premise is that, with clear and objective assessment procedures validated by an independent third party, it is not only possible to expand the reach of impact-oriented investments, but also develop new ways to fund and assess these projects. Furthermore, the lessons learned from these projects and assessments, documented in the form of rigorous research and case studies, can be used to expand our knowledge of how to improve and support the practical implementation and evaluation of high-impact projects.

RESEARCH **SUPERVISORS**

Sérgio Lazzarini (Insper) & Armen Ovanessoff (Accenture Research)

CORE PROJECT TEAM **AND RESEARCH FIELDWORK** (BRAZIL)

Nobuiuki Ito
Felippe Oliveira
Leandro Pongeluppe
Klayton Rocha

RESEARCH **FIELDWORK** (INDIA AND SOUTH AFRICA)

Marlene Joubert
Smriti Mathur
Udit Sabharwal

WE WOULD ALSO LIKE TO ACKNOWLEDGE THE FOLLOWING PEOPLE

FOR THEIR IMPORTANT CONTRIBUTIONS TO THIS RESEARCH:

Carolina Attili, Lance Ealey, Tharina Gombault, Karine Kirner, David Light, Raghav Narsalay, Paul Nunes, Marcel Pacano, Daniel Pacheco, Sergio Pedroso, Athena Peppes, Eduardo Plastino, Mark Purdy, Edson Saad, Yusof Seedat and Simone Silva.

OUR SPECIAL THANKS GO TO OUR SPONSORS

AT ACCENTURE HEALTH AND PUBLIC SERVICES:

Lourenço Mendonça & Carlos Pedranzini



LIST OF REFERENCES

¹ Olson, Lynn. Growing Pains. Education Week, 2 November 1994.

² Bradach, Jeffrey L. Going to Scale. Stanford Social Innovation Review, in Spring 2003. Available at: https://ssir.org/articles/entry/going_to_scale

²Júlio César Alexandre, Municipal Secretary of Education in Sobral – Interview to the GGN portal. Available at: <http://jornalggn.com.br/blog/luisnassif/a-experiencia-de-sobral-na-educacao>. Retrieved on December 1, 2016

³Júlio César Alexandre, Municipal Secretary of Education in Sobral – Presentation on Educational Policies to the Chamber of Deputies. Available at: <http://www2.camara.leg.br/atividade-legislativa/comissoes/comissoes-permanentes/ce/audiencias-publicas-1/apresentacoes/apresentacao-secretario-julio-cesar>. Retrieved on December 1, 2016

⁴Rocha, Komatsu & Menezes-Filho (2015) Sobral: um caso de sucesso na educação básica brasileira

⁵Osasco Municipality – Secretary of Housing and Urban Development Report (SEHDU) 2005-2008

⁶Osasco Municipality – Department of Housing and Urban Development: 10 years of Habitational Policies (2015). Available at <http://www4.pucsp.br/cedepe/download/apresentacoes/sandra-simoes.pdf>. Retrieved from November 29, 2016

⁷Chamber of deputies – Commission of Urban Development – 4th Citizen City Seal Award (2007). Available at <http://www2.camara.leg.br/camaranoticias/noticias/115783.html>. Retrieved at November 29, 2016

⁸Moura, M., Piza, P. And Poplawski-Ribeiro, M. The Distributive Effects of Land Title on Labor Supply: Evidence from Brazil (2011). International Monetary Fund Working Paper

⁹Associação Nacional das Empresas de Transportes Urbanos (NTU) (2012) Estudos de BRT no Brasil. Available at: <http://www.ntu.org.br/novo/upload/Publicacao/Pub635109526781748662.pdf> Retrieved on December 21, 2016



¹⁰Friberg, Lars. “Innovative Solutions for Public Transport; Curitiba, Brazil.” Sustainable Development International (2001): 153- 56. Available at: <http://infohouse.p2ric.org/ref/40/39732.pdf>. Retrieved on December 21, 2016.

¹¹Dados estatísticos (Statistical data). Available at: <https://www.poupatempo.sp.gov.br/>. Retrieved on December 19, 2016.

¹²Mobile Time [por Henrique Medeiro]. Poupatempo no celular: acesso patrocinado custará R\$ 20 milhões ao ano para o Estado de SP. Available at: <http://www.mobiletime.com.br/28/09/2015/poupatempo-no-celular-acesso-patrocinado-custara-r-20-milhoes-ao-ano-para-o-estado-de-sp/420765/news.aspx>. Retrieved on December 19, 2016.

¹³Literates and Literacy Rates - Census Provisional (2001). Available at http://www.nlm.nic.in/literacy01_nlm.htm. Retrieved on December 12, 2016

¹⁴National University of Education Planning and Administration, “District Report Cards” (2001). Available at: <http://www.dise.in/Downloads/Publications/Publications%202001-02/DRC2002/AndhraPradesh/rptNizamabad.htm> Retrieved on December 12, 2016

¹⁵Muralidharan, K., Sundararaman, V. (2009) TEACHER PERFORMANCE PAY: EXPERIMENTAL EVIDENCE FROM INDIA NBER WORKING PAPER SERIES - Working Paper 15323 - Available at: <http://www.nber.org/papers/w15323.pdf>. Retrieved on December 13, 2016

¹⁶Making schools work: new evidence on accountability reforms / Barbara Bruns, Deon Filmer, Harry Anthony Patrinos. ISBN: 978-0-8213-8679-8, eISBN: 978-0-8213-8680-4, DOI: 10.1596/978-0-8213-8679-8

¹⁷HABITAT, 1996. Managing Water Resources for Large Cities and Towns - Organisational Re-engineering for Capacity Development: A Case Study of Hyderabad Metropolitan Water Supply and Sewerage Board, India. Available at: <http://www.nzdl.org/gsdImod?e=d-00000-00---off-0cdl--00-0---0-10-0---0---Odirect-10---4-----0-1l--11-en-50---20-about---00-0-1-00-0--4----0-0-11-10-OutfZz-8-00&a=d&c=cdl&cl=CL1.45&d=HASH01808c12ae8210c8f518b9b3.6.14> Retrieved on January 16, 2017

¹⁸Kamalanathan, C.R. (n.d.). Improving Urban Water Supply and Sewerage Services: The Hyderabad Experiment. Available at: <http://unpan1.un.org/intradoc/groups/public/documents/cgg/unpan045230.pdf> Retrieved on December 28, 2016.

¹⁹Caseley, Jonathan (2003) Blocked drains and open minds: multiple accountability relationships and improved service delivery performance in an Indian city. IDS Working Paper 211. Available at: <http://unpan1.un.org/intradoc/groups/public/documents/other/unpan022458.pdf> Retrieved on December 28, 2016.

²⁰Water and Sanitation Program South Asia (WSP-SA) (n.d.). Benchmarking Urban Water Utilities in India. Available at: <http://documents.worldbank.org/curated/en/631341468281674829/pdf/470570WSP0Box31Benchmarking11Report.pdf> Retrieved on December 29, 2016.

²¹Directorate of Economics and Statistics (2014) - Statistical abstract of Delhi. Available at: http://www.delhi.gov.in/wps/wcm/connect/DOIT_DES/des/home Retrieved on December 28, 2016

²²Delhi Metro Rail Corporation Limited (1998). Annual Report 97-98. Available at: <http://www.delhimetrorail.com/OtherDocuments/AnnualReports/1997-98.pdf> Retrieved on December 28, 2016

²³Delhi Metro Rail Corporation (2016), Introduction. Available at: http://www.delhimetrorail.com/about_us.aspx#Introduction Retrieved on April 7, 2016

²⁴Delhi Metro Rail. Annual Report 2015-15. Available at: http://www.delhimetrorail.com/annual_report.aspx. Retrieved on December 27, 2016

²⁵Ramachandran, M. (2012). Metro rail projects in India: a study in project planning. Oxford University Press.

²⁶The Times India (2014). Delhi Metro rides high, keeps 3.9 lakh vehicles off roads. Available at: <http://timesofindia.indiatimes.com/city/delhi/Delhi-Metro-rides-high-keeps-3-9-lakh-vehicles-off-roads/articleshowprint/45636132.cms?null> Retrieved on January 3, 2017.

²⁷The Economist (2008). E for express Available at: http://www.economist.com/node/10638184/print?story_id=10638184 Retrieved on December 27, 2016

²⁹The Institute for Public-Private Partnerships, 2009. Public-Private Partnerships in E-Government: Handbook. Washington, DC: infoDev / World Bank. Available at <http://www.infodev.org/publications> Retrieved on December 27, 2016

³⁰Government of Telangana, “Mee Seva Authorized Counters”. Available at: <http://tgdept.meeSeva.gov.in/APSDCDeptPortal/UserInterface/Admin/MeeSevaCentresList.aspx> Retrieved on July 22, 2015

³¹IIM- Ahmedabad (2007), “Impact Assesment of E-government Projects in India”. Available at <http://www.iimahd.ernet.in/egov/documents/impact-assessment-study-dit.pdf> Retrieved on December 27, 2016

³²Prahalad, C. K. (2009). The fortune at the bottom of the pyramid, revised and updated 5th anniversary edition: Eradicating poverty through profits. FT Press.



³³South African National HIV Prevalence, Incidence and Behaviour Survey, 2012. Available at: <http://www.hsrc.ac.za/uploads/pageContent/4565/SABSSM%20IV%20LEO%20final.pdf> Retrieved on December 14, 2016

³⁴“Siyakha Nentsha” Enhancing the Economic, Health, and Social Capabilities of Highly Vulnerable Youth, by Kelly Hallman, Kasthuri Govender, Eva Roca, Emmanuel Mbatha and Mike Rogan. Available at: <http://poppov.org/~media/PopPov/Documents/events/4thAnnConf/hallman.ashx>. Retrieved on December 14, 2016

³⁵Hallman, M Calderon, K Govender, E Roca, E Mbatha, M Rogan, H Taboada, S Siema, J Faults, R Brown; Siyakha Nentsha: An experiment to build Social, Health and Financial capabilities of vulnerable girls and boys in KwaZulu-Natal, South Africa; Making Cents Conf. – 12 Sept. 2012; Population Council.

³⁶Kelly Hallman, Eva Roca, Kasthuri Govender, Emmanuel Mbatha, Rob Pattman, Deevia Bhana and Mike Rogan; Protection for the lifecourse: Enhancing health, social and economic capabilities of highly vulnerable adolescents; IUSSP; Marrakech, Morocco; September 28, 2009; Population Council.

³⁷Kelly Hallman and Eva Roca; Siyakha Nentsha: Building economic, health, and social capabilities among highly vulnerable adolescents in KwaZulu-Natal, South Africa; May 2011; Population Council. http://www.popcouncil.org/uploads/pdfs/TABriefs/04_SiyakhaNentsha.pdf

³⁸The Housing Development Agency (2012). KwaZulu-Natal: Informal settlements status

³⁹Managing eThekweni Water and Sanitation, Neil Macleod, Civil Engineering, October 2013.

⁴⁰CONSTITUTION OF THE REPUBLIC OF SOUTH AFRICA NO. 108 OF 1996. Available at <http://www.gov.za/sites/www.gov.za/files/images/a108-96.pdf>

⁴¹Hemson, D., Dube, B. Water services and public health: the 2000-01 cholera outbreak in KwaZulu-Natal, South Africa

⁴²PRESS RELEASE: “Most Progressive Water Utility in Africa” Wins 2014 Stockholm Industry Water Award via Dow Jones Factiva, Platts Commodity News, 24 May 2014.

⁴³eThekweni Water – Non-Revenue Water Update, February 2015

⁴⁴MyCiTi a game-changer for weary commuters. Available at: <http://www.news24.com/SouthAfrica/News/MyCity-a-game-changer-for-weary-commuters-20150601>. Retrieved on December 5, 2016

⁴⁵Integrated rapid transport: is the city of Cape Town utilising its full potential?. Available at: <http://www.netd.ac.za/?action=view&identifier=oai%3Aunion.ndltd.org%3Anwu%2Foai%3Adspace.nwu.ac.za%3A10394%2F4417>. Retrieved on December 7, 2016

⁴⁶Cape Town unveils 2032 transport plan
<http://futurecapetown.com/2015/06/cape-town-reveals-its-new-transportation-plan-the-2032-integrated-public-transport-network/#.WEW-jPkrKM8>. Retrieved on December 5, 2016

⁴⁷Myciti bus rapid transit it is not just about the bus. Available at: http://repository.up.ac.za/bitstream/handle/2263/45541/Bulman_Myciti_2014.pdf?sequence=1&isAllowed=y Retrieved on December 7, 2016.

⁴⁸MyCiTi innovations. Available at: <http://search.ebscohost.com/login.aspx?direct=true&db=plh&AN=82573519&site=ehost-live>

⁴⁹AWARDS AND ACCOLADES – Transport for Cape Town (2016). Available at: <http://www.tct.gov.za/en/awards/>. Retrieved on December 16, 2016.

⁵⁰Chidi, S. C. (2011). An evaluation of rural-urban migration and its link to informal settlement pattern: a case study of Disteneng in Polokwane Municipality in Limpopo Province (Doctoral dissertation, University of Limpopo - Turfloop Campus).

⁵¹Agarwal, R., Barney, J. B., Foss, N. J., & Klein, P. G. (2009). Heterogeneous resources and the financial crisis: implications of strategic management theory. *Strategic Organization*, 7(4): 467-484.

⁵²Kivleniece, I., & Quelin, B. V. 2012. Creating and capturing value in public-private ties: A private actor’s perspective. *Academy of Management Review*, 37(2): 272-299

⁵³Moore, M. H. 1995. *Creating public value: Strategic management in government*. Cambridge: Harvard University Press.

⁵⁴Kroeger, A., & Weber, C. (2014). Developing a conceptual framework for comparing social value creation. *Academy of Management Review*, 39(4): 513-540

⁵⁵Moskal, B. M. (2000). Scoring rubrics: What, when and how? *Practical Assessment, Research & Evaluation*, 7(3).

⁵⁶Oakleaf, M. (2009). Using rubrics to assess information literacy: An examination of methodology and interrater reliability. *Journal of the American Society for Information Science and Technology*, 60(5): 969-983.

⁵⁷Fleiss, J. L., Levin, B., & Paik, M. C. (2003). *Statistical methods for rates and proportions*. New York: Wiley.



